Unit-3

Network Layer

- Network layer Design issues
- **Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast routing, Multicast routing, distance vector routing**
- Congestion Control Algorithms
- Quality of Service
- Internetworking
- The Network layer in the internet.

Introduction of Network layer

- The network layer or layer 3 of the OSI (Open Systems Interconnection) model is concerned delivery of data packets from the source to the destination across multiple links.
- The designers who are concerned with designing this layer needs to cater to certain issues. These issues encompasses **the services provided to the upper layers as well as internal design of the layer.**
- **Before learning about design issues in the network layer, let's learn about it's various functions.**

1. Addressing:

Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in network.

2. Packeting:

This is performed by Internet Protocol. The network layer converts the packets from its upper layer.

3. Routing:

It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.

4. Inter-networking:

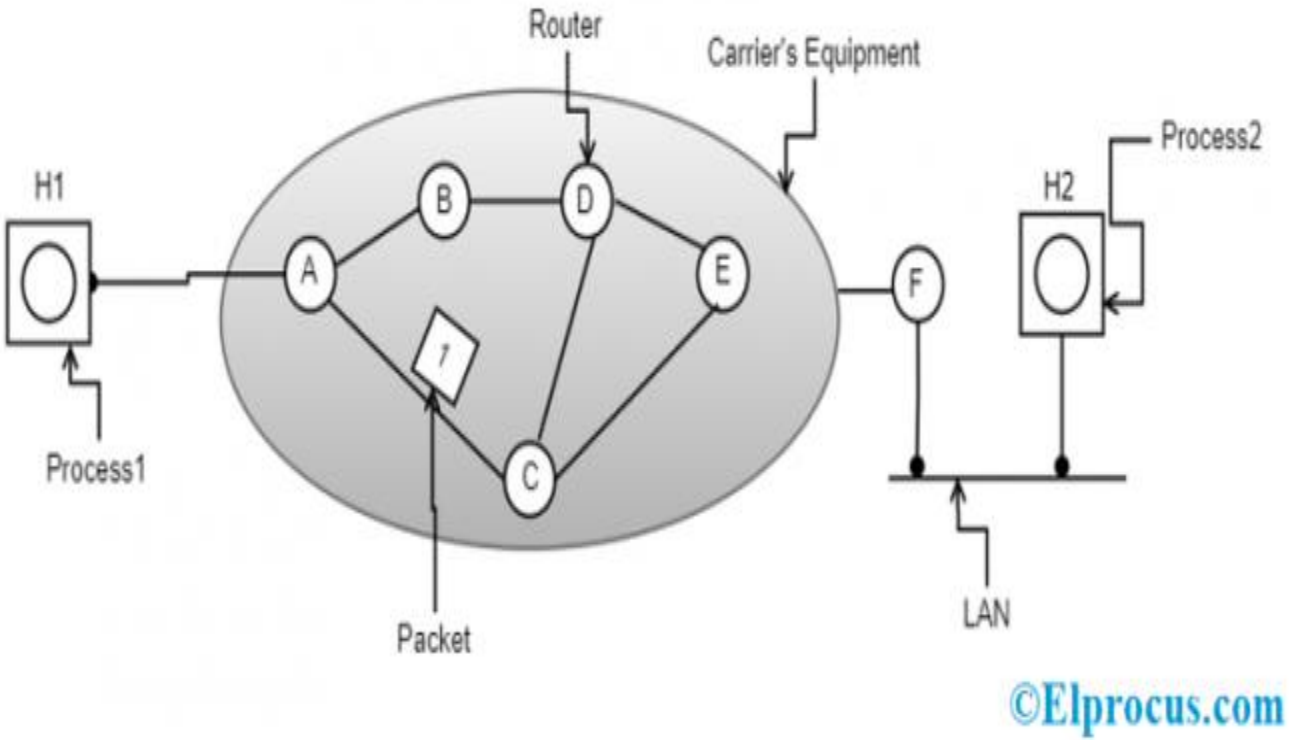
It works to deliver a logical connection across multiple devices.

NETWORK LAYER DESIGN ISSUES

- The network layer comes with some design issues they are described as follows:

1. Store and Forward packet switching:

- The network layer operates in an environment that uses store and forward packet switching.
- The node which has a packet to send, delivers it to the nearest router. **The packet is stored in the router until it has fully arrived and its checksum is verified for error detection.** Once, this is done, the packet is forwarded to the next router.
- **Since, each router needs to store the entire packet before it can forward it to the next hop/link, the mechanism is called store − and − forward switching.**



Here, the foremost elements are the carrier's equipment (the connection between routers through transmission lines) and the customer's equipment.

- H1 has a direct connection with carrier router ‘A’, while H2 is connected to carrier router ‘F’ on a LAN connection.
- One of the carrier router ‘F’, is pointed outside the carrier’s equipment as it does not come under the carrier, whereas considered as protocols, software, and construction.
- This switching network performs as Transmission of data happens when the host (H1) with a packet transfers it to the nearby router through LAN (or) point-to-point connection to the carrier. The carrier stores the packet until it completely arrives thus confirms the checksum.
- Then after, the packet is transmitted over the path until H2 is reached.

2. Services provided to Transport Layer

- The network layer provides service its immediate upper layer, namely transport layer, through the network − transport layer interface. The two types of services provided are −
  - **Connection − Oriented Service** − In this service, a path is setup between the source and the destination, and all the data packets belonging to a message are routed along this path.
  - **Connectionless Service** − In this service, each packet of the message is considered as an independent entity and is individually routed from the source to the destination.

The objectives of the network layer while providing these services are −

i. The services should not be dependent upon the router technology.

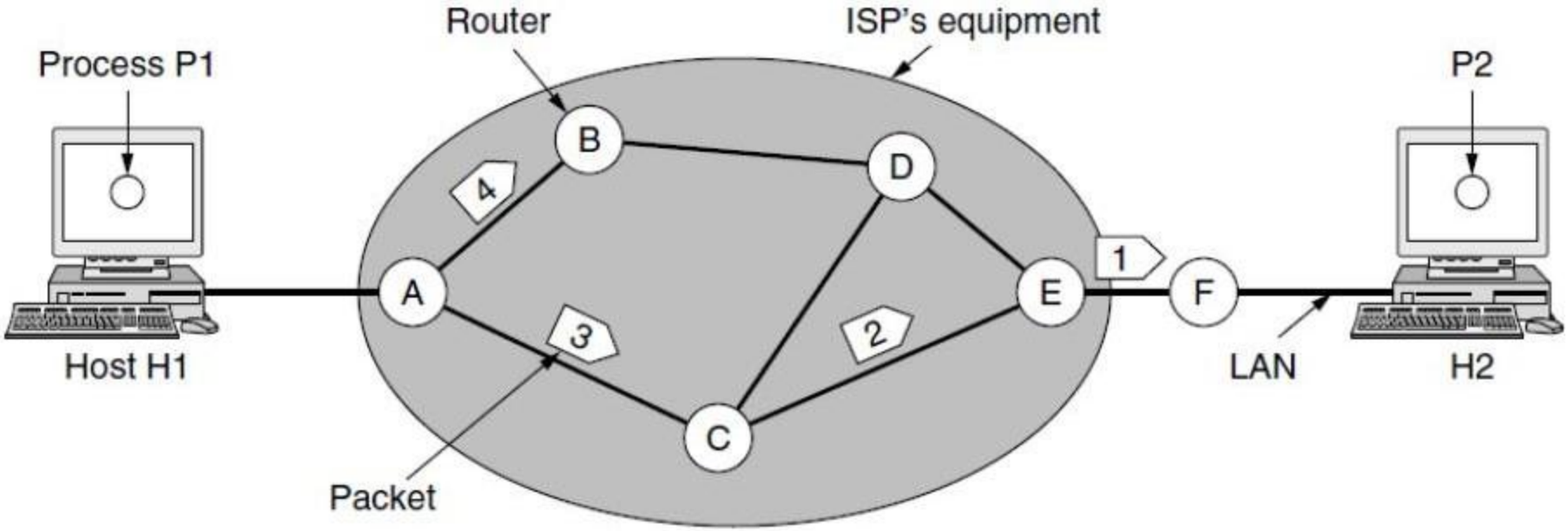
ii. The transport layer should be shielded from the number, type, topology of the routers present.

iii. A uniform addressing plan should be made available to the transport layer, whether the network is a LAN, MAN or WAN.

3. Implementation of Connectionless Service

- In connectionless service, each packet is transmitted independently, each packet contains its routing information and is **termed as datagram**.
- The network using datagrams for transmission is called **datagram networks or datagram subnets**.
- **No prior setup of routes are needed before transmitting a message**. Each datagram belong to the message follows its own individual route from the source to the destination.
- **An example of connectionless service is Internet Protocol or IP.**

- When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and then transmits each packet to router ‘A’ through a few protocols.
- **Each router is provided with a routing table where it decides the destination points.**
  In the below figure, it is clear that packets from ‘A’ need to be transmitted either to B or C even when the destination is ‘F’. The routing table of ‘A’ is clearly outlined above.



A's table (initially)

Dest.	Line
A	–
B	B
C	C
D	B
E	C
F	C

A's table (later)

A	–
B	B
C	C
D	B
E	B
F	B

C's table

A	A
B	A
C	–
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	–
F	F

Routing within a datagram network.

- Whereas in the case of packet 4, the packet from ‘A’ is routed to ‘B’, even the destination node is ‘F’. Packet ‘A’ chooses to transmit packet 4 through a different path than the initial three paths. This might happen because of traffic congestion along the path ACE.

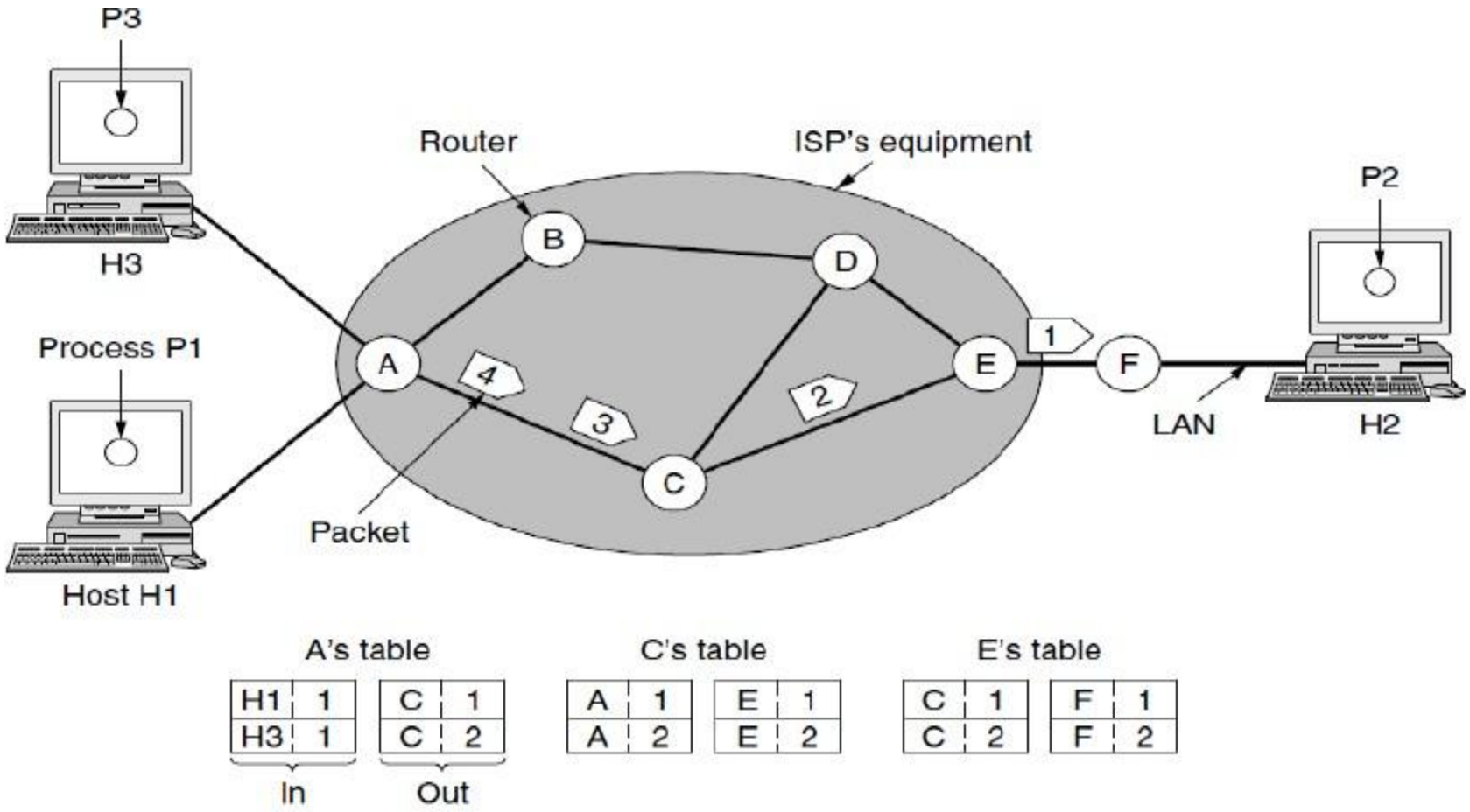
4). Implementation of Connection-Oriented Service

- To use a connection-oriented service, first we establishes a connection, use it and then release it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender.

It can be done in either two ways :

- **Circuit Switched Connection –** A dedicated physical path or a circuit is established between the communicating nodes and then data stream is transferred.

- **Virtual Circuit Switched Connection –** The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While, other connections may also be using the same path.



Routing within a virtual-circuit network.

- consider the situation shown in Fig. Here, host *H1* has established connection 1 with host *H2*. This connection is remembered as the first entry in each of the routing tables. The first line of *A*'s table says that if a packet bearing connection identifier 1 comes in from *H1*, it is to be sent to router *C* and given connection identifier 1. Similarly, the first entry at *C* routes the packet to *E*, also with connection identifier 1.
- Now let us consider what happens if *H3* also wants to establish a connection to *H2*. It chooses connection identifier 1 (because it is initiating the connection and this is its only connection) and tells the network to establish the virtual circuit. This leads to the second row in the tables.
- Note that we have a conflict here because although *A* can easily distinguish connection 1 packets from *H1* from connection 1 packets from *H3*, *C* cannot do this. For this reason, *A* assigns a different connection identifier to the outgoing traffic for the second connection.

- Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets. In some contexts, this process is called **label switching**.
- An example of a connection-oriented network service is **MPLS** (**Multi Protocol Label Switching**). It is used within ISP networks in the Internet, with IP packets wrapped in an MPLS header having a 20-bit connection identifier or label. MPLS is often hidden from customers, with the ISP establishing long-term connections for large amounts of traffic, but it is increasingly being used to help when quality of service is important but also with other ISP traffic management tasks.

Routing algorithms

Introduction of Routing algorithms

- In order to transfer the packets from source to the destination, the network layer must determine the best route through which packets can be transmitted.
- Whether the network layer provides datagram service or virtual circuit service, the main job of the network layer is to provide the best route. The routing protocol provides this job.
- The routing protocol is a routing algorithm that provides the best path from the source to the destination. The best path is the path that has the "least-cost path" from source to the destination.
- Routing is the process of forwarding the packets from source to the destination but the best route to send the packets is determined by the routing algorithm.

Classification of a Routing algorithm

- The Routing algorithm is divided into two categories:
    1. Adaptive Routing algorithm or dynamic routing
    2. Non-adaptive Routing algorithm or static routing

1. Adaptive Routing algorithm or dynamic routing

- This algorithm makes the routing decisions based on the topology and network traffic.
- The main parameters related to this algorithm are hop/links count, distance and estimated transit time.

An adaptive routing algorithm can be classified into three parts:

i. **Centralized algorithm:** It is also known as **global routing algorithm** as it computes the least-cost path between source and destination by using complete and global knowledge about the network.

- This algorithm takes the connectivity between the nodes and link cost as input, and this information is obtained before actually performing any calculation. **Link state algorithm** is referred to as a centralized algorithm since it is aware of the cost of each link in the network.

ii. Isolation algorithm: It is an algorithm that obtains the routing information by using local information rather than gathering information from other nodes.

iii. Distributed algorithm: It is **also known as decentralized algorithm** as it computes the least-cost path between source and destination in an iterative and distributed manner.

- In the decentralized algorithm, no node has the knowledge about the cost of all the network links.
- In the beginning, a node contains the information only about its own directly attached links and through an iterative process of calculation computes the least-cost path to the destination. **A Distance vector algorithm is a decentralized algorithm** as it never knows the complete path from source to the destination, instead it knows the direction through which the packet is to be forwarded along with the least cost path.

2. Non-adaptive Routing algorithm or static routing

- When booting up the network, the routing information stores to the routers.
- Non Adaptive routing algorithms do not take the routing decision based on the network topology or network traffic.

The Non-Adaptive Routing algorithm is of two types:

i. **Flooding:** In case of flooding, every incoming packet is sent to all the outgoing links except the one from it has been reached. **The disadvantage of flooding is that node may contain several copies of a particular packet.**

ii. **Random walks:** In case of random walks, a packet sent by the node to one of its neighbors randomly. An advantage of using random walks is that it uses the alternative routes very efficiently.

Differences b/w Adaptive and Non-Adaptive Routing Algorithm

Basis Of Comparison	Adaptive Routing algorithm	Non-Adaptive Routing algorithm
Define	Adaptive Routing algorithm is an algorithm that constructs the routing table based on the network conditions.	The Non-Adaptive Routing algorithm is an algorithm that constructs the static table to determine which node to send the packet.
Usage	Adaptive routing algorithm is used by dynamic routing.	The Non-Adaptive Routing algorithm is used by static routing.
Routing decision	Routing decisions are made based on topology and network traffic.	Routing decisions are the static tables.
Categorization	The types of adaptive routing algorithm, are Centralized, isolation and distributed algorithm.	The types of Non Adaptive routing algorithm are flooding and random walks.
Complexity	Adaptive Routing algorithms are more complex.	Non-Adaptive Routing algorithms are simple.

shortest path routing

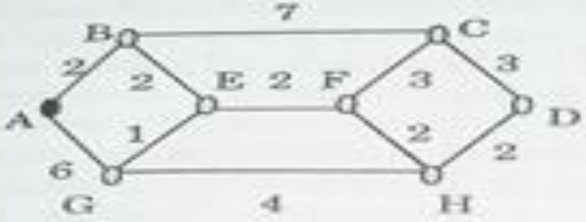
- It is one of the simple static routing algorithms that widely used for routing in the network.
- The basic idea is to build a graph with each node representing a router and each line representing a communication link. To choose a route between any two nodes in the graph, the algorithm finds the shortest path between the nodes.
- common shortest path algorithm is **Dijkstra Algorithm**

Dijkstra Algorithm

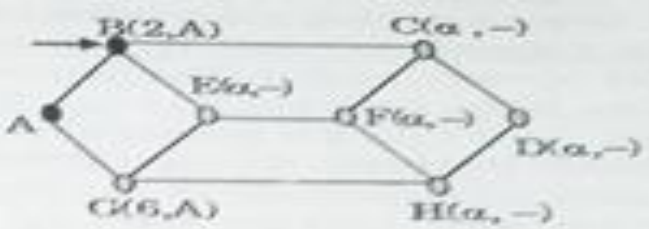
- The shortest path means the path in which anyone or more metrics are minimized. The metric may be distance, bandwidth, average traffic, communication cost, mean queue length, measured delay or any other factor.
- **Dijkstra Algorithm is mostly used for this purpose. The basic objective of the algorithm is to label each node with its distance from the source node along the best-known path.**

Steps of Dijkstra Shortest Path Routing Algorithm:

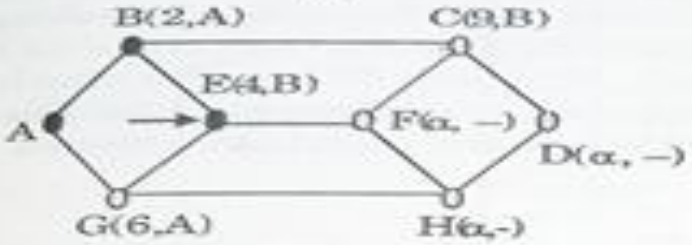
1. Initially, no path is known as – All nodes are labelled with infinity.
2. As the algorithm progresses labels change reflecting the better path. The label called as **tentative label**.
3. As it is found that the label represents the shortest path then it is made **permanent** and never changed thereafter.


B
C
7
2
2
E 2 F
3
3
A
D
1
2
2
6
G
4
H

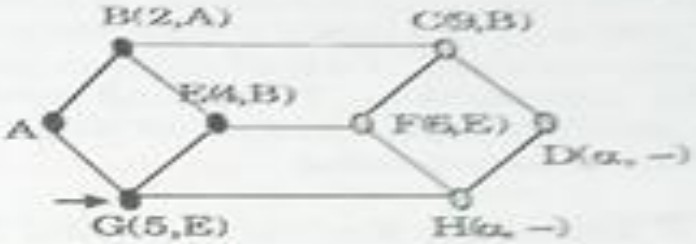
(a)


B(2,A)
C(α, –)
E(α,–)
A
F(α,–)
D(α,–)
G(6,A)
H(α, –)

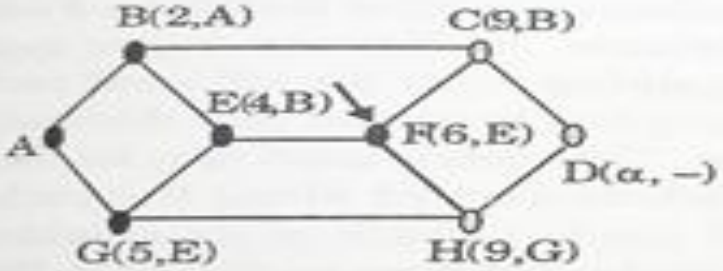
(b)


B(2,A)
C(9,B)
E(4,B)
A
F(α, –)
D(α, –)
G(6,A)
H(α,-)

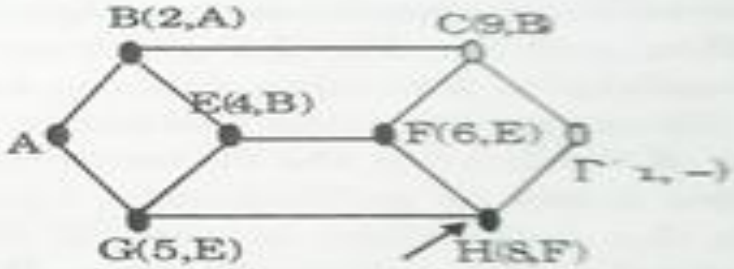
(c)


B(2,A)
C(9,B)
E(4,B)
A
F(6,E)
D(α, –)
G(5,E)
H(α, –)

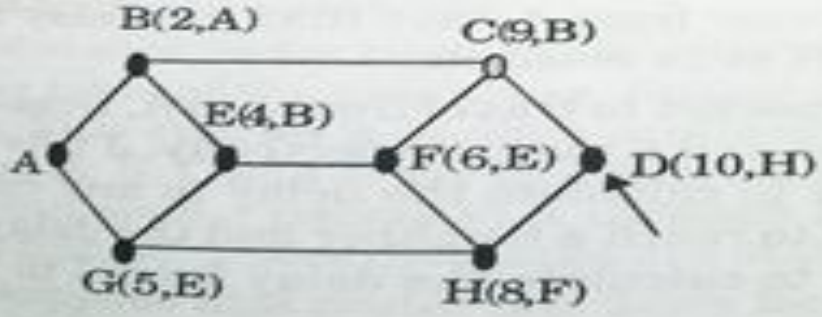
(d)


B(2,A)
C(9,B)
E(4,B)
A
F(6,E)
D(α, –)
G(5,E)
H(9,G)

(e)

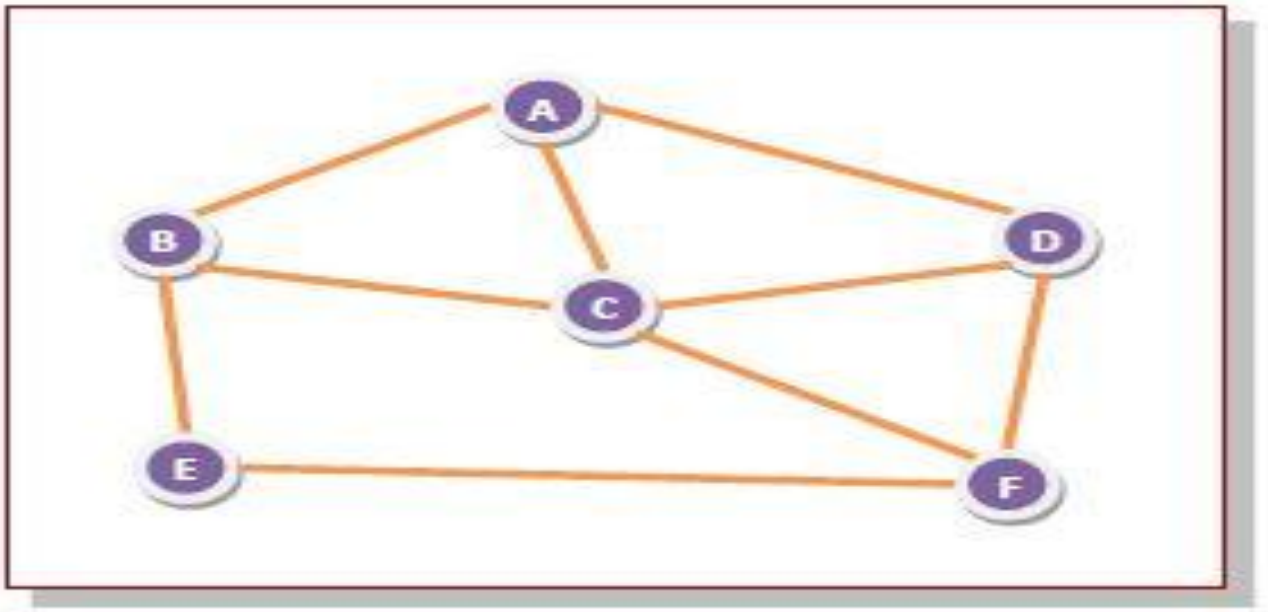

B(2,A)
C(9,B
E(4,B)
A
F(6,E)
G(5,E)
H(8,F)

(f)


B(2,A)
C(9,B)
E(4,B)
A
F(6,E)
D(10,H)
G(5,E)
H(8,F)

Flooding routing algorithm

- Flooding is a **non-adaptive routing technique** following this simple method: when a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.
- For example, let us consider the network in the figure, having six routers that are connected through transmission lines.



Using flooding technique −

- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C and E.

Types of Flooding

Flooding may be of three types −

- **Uncontrolled flooding** − Here, each router unconditionally transmits the incoming data packets to all its neighbours.
- **Controlled flooding** − They use some methods to control the transmission of packets to the neighbouring nodes. The two popular algorithms for controlled flooding are **Sequence Number Controlled Flooding (SNCF) and Reverse Path Forwarding (RPF).**
- **Selective flooding** − Here, the routers don't transmit the incoming packets only along those paths which are heading towards approximately in the right direction, instead of every available paths.

Advantages of Flooding

- It is very simple to setup and implement, since a router may know only its neighbours.
- It is extremely robust. Even in case of malfunctioning of a large number routers, the packets find a way to reach the destination.
- All nodes which are directly or indirectly connected or visited. So, there are no chances for any node to be left out. This is a main criteria in case of broadcast messages.
- The shortest path is always chosen by flooding.

Disadvantages

- Flooding generates a vast number of duplicate packets
- Suitable damping mechanism must be used

Hop-Count

- A hop counter may be contained in the packet header which is decremented at each hop.
  with the packet being discarded when the counter becomes zero ,The sender initializes the hop counter. If no estimate is known, it is set to the full diameter of the subnet.

sequence number

- Keep track of the packets which are responsible for flooding using **a sequence number**. Avoid sending them out a second time.

Hierarchical routing

- In hierarchical routing, the **routers are divided into regions**.
- Each router has complete details about how to route packets to destinations within its own region. But it does not have any idea about the internal structure of other regions.
- Every router needs to save some information about other routers. When network size is growing, the number of routers in the network will increase. Therefore, the size of routing table increases, then routers cannot handle network traffic as efficiently. To overcome this problem we are using hierarchical routing.

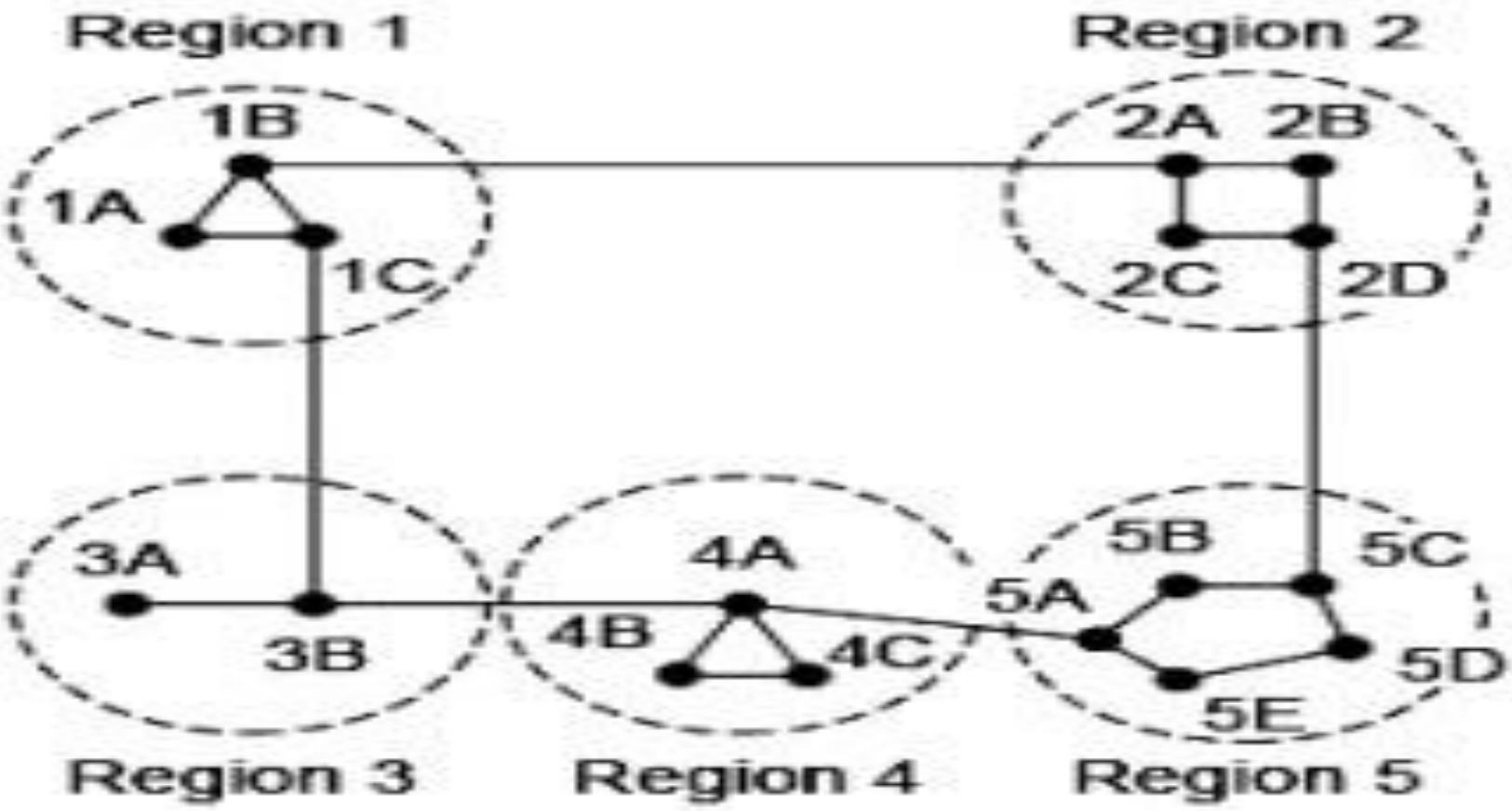
- In hierarchical routing, **routers are classified in groups called regions**. Each router has information about the routers in its own region and it has no information about routers in other regions. So, routers save one record in their table for every other region.
- For huge networks, a two-level hierarchy may be insufficient hence, it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups and so on.

- **Gateway is a router that knows about other regions**

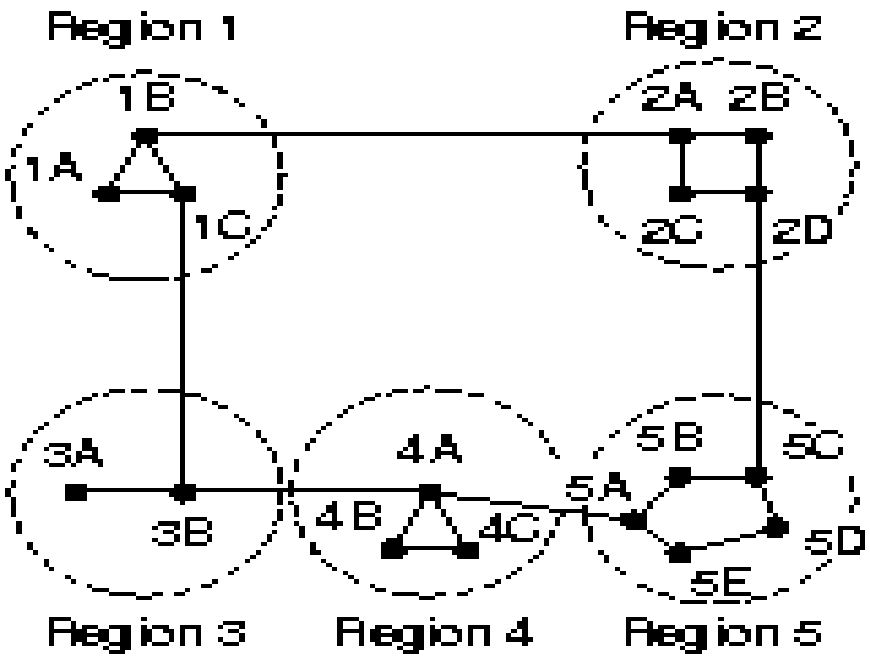
- **The main advantage of the hierarchical routing is scalability. It shows the capability of the router when the specific operations are performed on the network.**

Example

- Consider an example of two-level hierarchy with five regions as shown in figure −



- Let see the full routing table for router 1A which has 17 entries, as shown below −



Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

Hierarchical table

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Explanation

- **Step 1** − For example, the best path from 1A to 5C is via region 2, but hierarchical routing of all traffic to region 5 goes via region 3 as it is better for most of the other destinations of region 5.
- **Step 2** − Consider a subnet of 720 routers. If no hierarchy is used, each router will have 720 entries in its routing table.
- **Step 3** − Now if the subnet is partitioned into 24 regions of 30 routers each, then each router will require 30 local entries and 23 remote entries for a total of 53 entries.

Broadcast routing

- Sending a packet to all the nodes on the network simultaneously is called **broadcasting**.
- In some applications, hosts need to send messages to many or all other hosts. **For example, a service distributing weather reports, stock market updates, or live radio programs** might work best by broadcasting to all machines and letting those that are interested read the data. various methods have been proposed for doing it.

Methods:

i. Distinct point to point counting:

- This is a simply send a distinct packet to each destination. Hence it is waste of bandwidth but it also require complete list of all destination.

ii. Flooding:

- flooding algorithm send packet on every outgoing line except the line on which it arrived.
- Drawback **more bandwidth** and **consumed duplicate packets.**.

iii. Multidestination routing

- In which each packet contains either a list of destinations or a bit map indicating the desired destinations. When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed.(An output line is needed if it is the best route to at least one of the destinations.)
- Based on that the router generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line.

Iv. Use of spanning tree : It makes explicit use of the sink tree—or any other convenient spanning tree for the router initiating the broadcast.

- A **spanning tree** is a subset of the network that includes all the routers but contains no loops.
- **Sink trees** are spanning trees. If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on.
- **This method makes excellent use of bandwidth, generating the absolute minimum number of packets necessary to do the job**.

v. Reverse Path Forwarding

- When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the line that is normally used for sending packets to the source of the broadcast.
- If so, there is an excellent chance that the broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router.
- This being the case, the router forwards copies of it onto all lines except the one it arrived on. If, however, the broadcast packet arrived on a line other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.

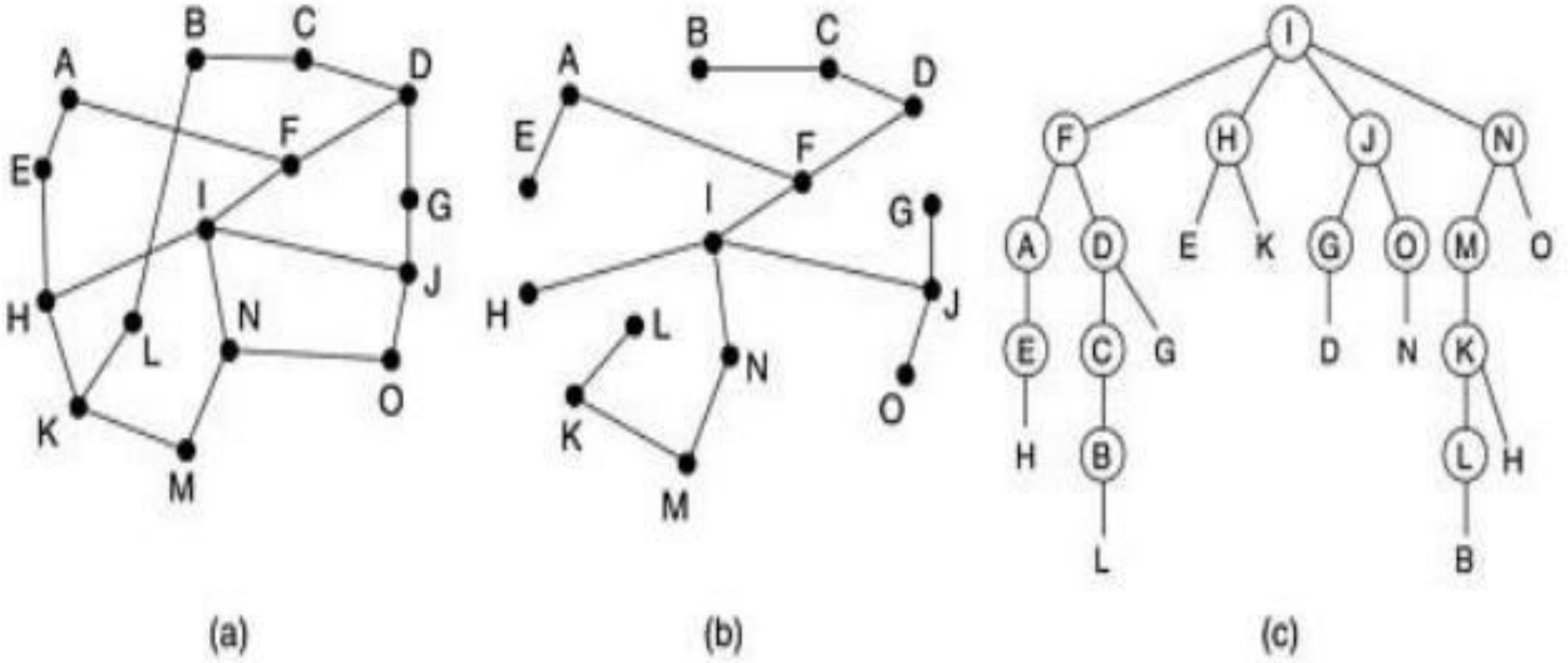


FIG: Reverse path forwarding. (a) A subnet. (b) A sink tree. (c) The tree built by reverse path forwarding.

- An example of reverse path forwarding is shown in Fig. Part (a) shows a subnet, part shows a sink tree for router I of that subnet, and part (c) shows how the reverse path algorithm works. On the first hop, I sends packets to F, H, J, and N, as indicated by the second row of the tree. Each of these packets arrives on the preferred path to I (assuming that the preferred path falls along the sink tree) and is so indicated by a circle around the letter. On the second hop, eight packets are generated, two by each of the routers that received a packet on the first hop.
- As it turns out, all eight of these arrive at previously unvisited routers, and five of these arrive along the preferred line. Of the six packets generated on the third hop, only three arrive on the preferred path (at C, E, and K); the others are duplicates. After five hops and 24 packets, the broadcasting terminates, compared with four hops and 14 packets had the sink tree been followed exactly.

- **The advantage of reverse path forwarding** is that it is both reasonably efficient and easy to implement.
- It does not require routers to know about spanning trees, nor does it have the overhead of a destination list or bit map in each broadcast packet as does multidestination addressing. Nor does it require any special mechanism to stop the process, as flooding does (either a hop counter in each packet and a priori knowledge of the subnet diameter, or a list of packets already seen per source).

➢ **When a device request to broadcast, it sends a packet to the network's broadcast address. Then a device such as a router or a switch transmits that packet to other devices in the network. All the devices that can receive these packets belong to a broadcast domain. However, one major drawback in broadcasting is that it can cause security issues and generate data loss.**

Multicast Routing Algorithm

- **Multicasting** in computer network **is a group communication**, where a sender(s) send data to multiple receivers simultaneously.
- It supports one – to – many and many – to – many data transmission across LANs or WANs.
- Some applications require that widely-separated processes work together in groups, for example, a group of processes implementing a distributed database system. In these situations, it is frequently necessary for one process to send a message to all the other members of the group. If the group is small, it can just send each other member a point-to-point message.

- If the group is large, this strategy is expensive. Sometimes broadcasting can be used, but using broadcasting to inform 1000 machines on a million-node network is inefficient because most receivers are not interested in the message **.**
- **Sending a message to such a group is called multicasting**, and **its routing algorithm is called multicast routing.**

- There are various multicasting based applications. **A user can send an email to some of the recipients rather than sending emails to all the recipients in the contact list.**
- Webcasts and video conferencing also use multicasting.
- Furthermore, devices such as routers and switches send packets to a group of devices in the network with the help of multicasting.

- **Multicasting requires group management. Some way is needed to create and destroy groups, and to allow processes to join and leave groups.**
- It is important that routers know which of their hosts belong to which groups. Either hosts must inform their routers about changes in group membership, or routers must query their hosts periodically.
- Either way, routers learn about which of their hosts are in which groups. Routers tell their neighbours, so the information propagates through the subnet. To do multicast routing, each router computes a spanning tree covering all other routers.

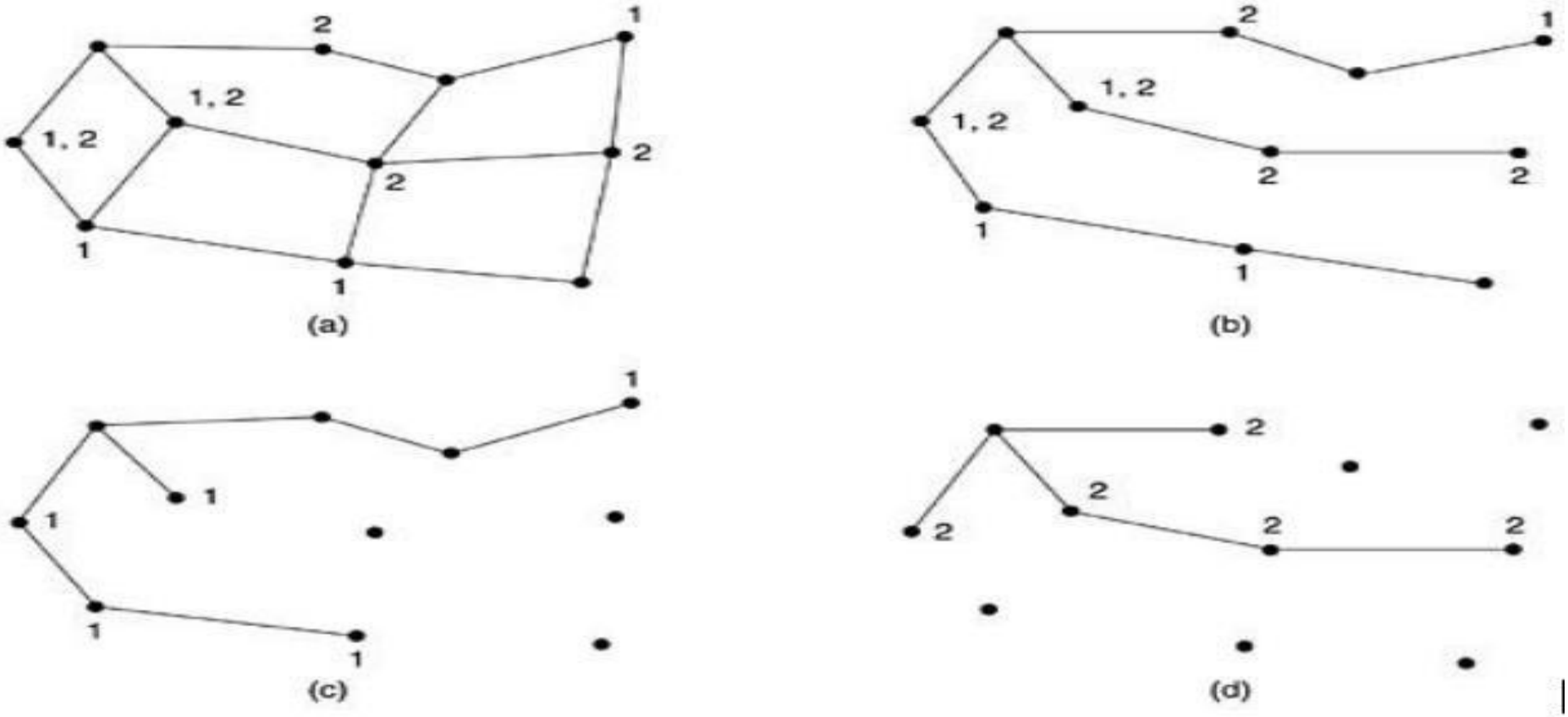


Fig. (a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

- For example, in Fig. (a) there are two groups, 1 and 2. Some routers are attached to hosts that belong to one or both of these groups, as indicated in the figure. A spanning tree for the leftmost router is shown in Fig. (b). When a process sends a multicast packet to a group, the first router examines its spanning tree and prunes it, removing all lines that do not lead to hosts that are members of the group.
- **In our example, Fig. (c) shows the pruned spanning tree for group 1.**
- **Similarly, Fig. (d) shows the pruned spanning tree for group 2. Multicast packets are forwarded only along the appropriate spanning tree.**

- Various ways of pruning the spanning tree are possible.
- The simplest one can be used if **link state routing** is used and each router is aware of the complete topology, including which hosts belong to which groups.
- Then the spanning tree can be pruned, starting at the end of each path, working toward the root, and removing all routers that do not belong to the group in question.
- With **distance vector routing**, a different pruning strategy can be followed.

- One **disadvantage of this algorithm is that it scales poorly to large networks**.
- Suppose that a network has n groups, each with an average of m members. For each group, m pruned spanning trees must be stored, for a total of mn trees.

DISTANCE VECTOR ROUTING(DV)

- Computer networks generally use dynamic routing algorithms that are **more complex than flooding, but more efficient because they find shortest paths for the current topology.** Two dynamic algorithms in particular, **distance vector routing and link state routing,** are the most popular.
- **A distance vector routing algorithm operates by having each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there.**
- These tables are updated by exchanging information with the neighbors. Eventually, every router knows the best link to reach each destination.

- The distance vector routing algorithm is sometimes called by other names, most commonly the **distributed Bellman-Ford routing algorithm**, after the researchers who developed it (Bellman, 1957; and **Ford and Fulkerson**, 1962).
- The Distance vector algorithm is a dynamic algorithm.
- It is mainly used in ARPANET, and internet.
- **Each router maintains a distance table known as vector**.
- **The Distance vector algorithm is iterative, asynchronous and distributed.**
- Let dx(y)be the least cost path from node x to node y. the least costs are related by **Bellman ford equation:**

$$d_x(y) = \min_v\{c(x,v) + d_v(y)\}$$

- **$d_x(y) = \min_v\{c(x,v) + d_v(y)\}$**

Here x is the source node

Y is the destination node

V is the intermediate node(neighbors node)

- **Where** the minv is the equation taken for all x neighbors. After traveling from x to v, if we consider the least-cost path from v to y, the path cost will be $c(x,v)+d_v(y)$. The least cost from x to y is the minimum of $c(x,v)+d_v(y)$ taken over all neighbors.

Explanation:

1. Router transmits its **distance vector to each of its neighbors.**

2. Each router receives and saves the most recently received **distance vector** from each of its neighbors.

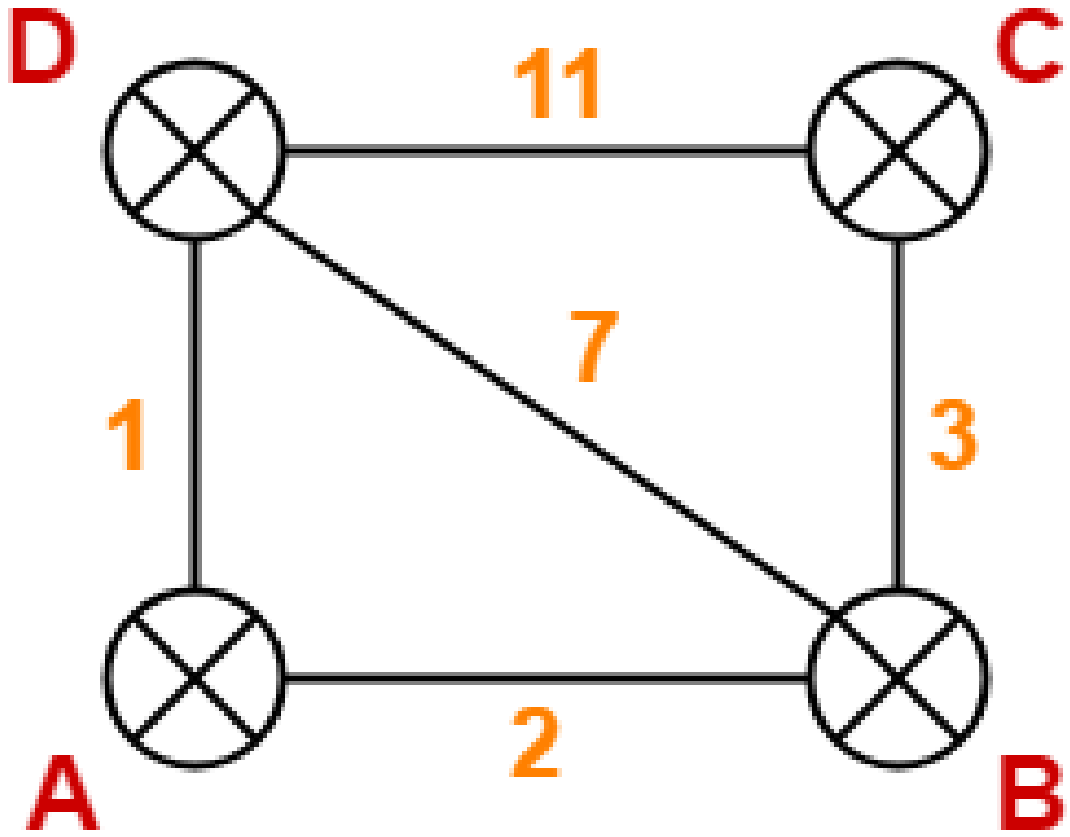
3. A router **recalculates** its distance vector when:

    a. It receives a distance vector from a neighbor containing different information than before.

    b. It discovers that a link to a neighbor has gone(i.e a topology change).

The Distance vector calculation based on minimizing the cost to each destination.

Example1



Each router prepares its routing table using its local knowledge.
Routing table prepared by each router is shown below-

At Router A-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	∞	–
D	1	D

- A->B distance

A->D->C->B=1+11+3=15

A->B=2

At Router B-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	7	D

At Router C-

Destination	Distance	Next Hop
A	∞	–
B	3	B
C	0	C
D	11	D

- **At Router D-**

Destination	Distance	Next Hop
A	1	A
B	7	B
C	11	C
D	0	D

- Each router exchanges its distance vector with its neighbors.
- After exchanging the distance vectors, each router prepares a new routing table.

- Now , Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-

From B

2
0
3
7

Cost(A→B) = 2

From D

1
7
11
0

Cost(A→D) = 1

New Routing Table at Router A

Destination	Distance	Next hop
A	0	A
B		
C		
D		

- Cost of reaching destination B from router A = min { 2+0 , 1+7 } = 2 via B.
- Cost of reaching destination C from router A = min { 2+3 , 1+11 } = 5 via B.
- Cost of reaching destination D from router A = min { 2+7 , 1+0 } = 1 via D.

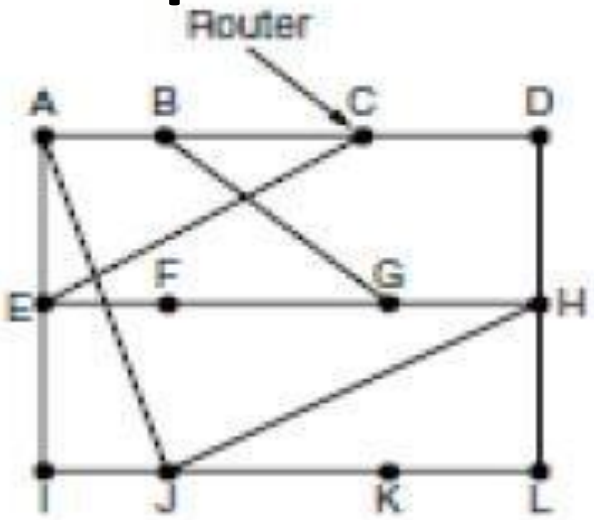
Explanation For Destination B

- Router A can reach the destination router B via its neighbor B or neighbor D.
- It chooses the path which gives the minimum cost.
- Cost of reaching router B from router A via neighbor B = Cost (A→B) + Cost (B→B)= **2 + 0** = 2
- Cost of reaching router B from router A via neighbor D = Cost (A→D) + Cost (D→B) = **1 + 7** = 8
- Since the cost is minimum via neighbor B, so router A chooses the path via B.
- It creates an entry (2, B) for destination B in its new routing table.
- Similarly, we calculate the shortest path distance to each destination router at every router.

Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

Example2



(a)

New estimated delay from J

To	A	I	H	K	Line
A	0	24	20	21	8 A
B	12	36	31	28	20 A
C	25	18	19	36	28 I
D	40	27	8	24	20 H
E	14	7	30	22	17 I
F	23	20	19	40	30 I
G	18	31	6	31	18 H
H	17	20	0	19	12 H
I	21	0	14	22	10 I
J	9	11	7	10	0 —
K	24	22	22	0	6 K
L	29	33	9	9	15 K
	JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6	New routing table for J

Vectors received from J's four neighbors

(b)

Advantages of Distance Vector routing –

- It is simpler to configure
- easy to implement and administer.

Disadvantages:

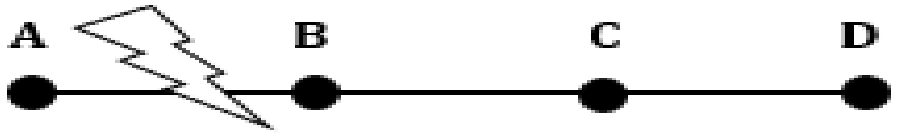
- Distance Vector Routing suffers from count to infinity problem.
- Distance Vector Routing uses UDP at transport layer.
- For larger networks, distance vector routing results in larger routing tables ,since each router must know about all other routers. This can also lead to congestion on WAN links.

Count to infinity problem:

1. One of the important issue in Distance Vector Routing is County of Infinity Problem.
2. Counting to infinity is just another name for a routing loop.
3. In distance vector routing, routing loops usually occur when an interface goes down.
4. It can also occur when two routers send updates to each other at the same time.

- Example:

Link Between A & B is Broken



	A	B	C	D
A	0, -	1, A	2, B	3, C
B	1, B	0, -	2, C	3, D
C	2, B	1, C	0, -	1, C
D	3, B	2, C	1, D	0, -

- Imagine a network with a graph as shown above in figure.
- As you see in this graph, there is only one link between A and the other parts of the network.

- Now imagine that the link between A and B is cut.
- At this time, B corrects its table.
- After a specific amount of time, routers exchange their tables, and so B receives C's routing table.
- Since C doesn't know what has happened to the link between A and B, it says that it has a link to A with the weight of 2 (1 for C to B, and 1 for B to A -- it doesn't know B has no link to A).
- B receives this table and thinks there is a separate link between C and A, so it corrects its table and changes infinity to 3 (1 for B to C, and 2 for C to A, as C said).
- Once again, routers exchange their tables.

- When C receives B's routing table, it sees that B has changed the weight of its link to A from 1 to 3, so C updates its table and changes the weight of the link to A to 4 (1 for C to B, and 3 for B to A, as B said).
- This process loops until all nodes find out that the weight of link to A is infinity.
- This situation is shown in the table below 4.2.
- In this way, Distance Vector Algorithms have a slow convergence rate.

	B	C	D
Sum of Weight to A after link cut	∞, A	2, B	3, C
Sum of Weight to A after 1st updating	3, C	2, B	3, C
Sum of Weight to A after 2nd updating	3, C	4, B	3, C
Sum of Weight to A after 3rd updating	5, C	4, B	5, C
Sum of Weight to A after 4th updating	5, C	6, B	5, C
Sum of Weight to A after 5th updating	7, C	6, B	7, C
Sum of Weight to A after nth updating	.....	....	....
∞	∞	∞	∞

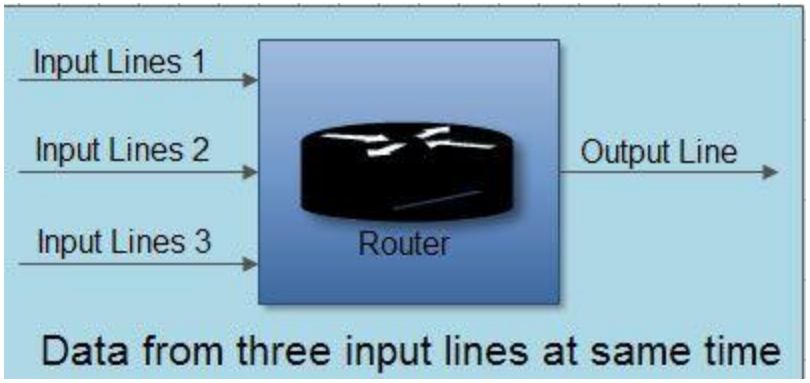
- One way to solve this problem is for routers to send information only to the neighbors that are not exclusive links to the destination.
- For example, in this case, C shouldn't send any information to B about A, because B is the only way to A.

Congestion Control Algorithms

- Congestion is a situation in Communication Networks in which too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called **congestion.**
- **Congestion in a network may occur when the load on the network *(i.e.* the number of packets sent to the network) is greater than the capacity of the network *(i.e.* the number of packets a network can handle.). Network congestion occurs in case of traffic overloading.**
- **The network and transport layers share the responsibility for handling congestion**. Since congestion occurs within the **network**, it is the network layer that directly experiences it and must ultimately determine what to do with the excess packets.
- However, the most effective way to control congestion is to reduce the load that the transport layer is placing on the network. This requires the network and transport layers to work together.

Causing of Congestion:

- The various causes of congestion in a subnet are:
- The input traffic rate exceeds the capacity of the output lines. If suddenly, a stream of packet start arriving on three or four input lines and all need the same output line. In this case, a queue will be built up. If there is insufficient memory to hold all the packets, the packet will be lost. Increasing the memory to unlimited size does not solve the problem. This is because, by the time packets reach front of the queue, they have already timed out (as they waited the queue).
- When timer goes off source transmits duplicate packet that are also added to the queue. Thus same packets are added again and again, increasing the load all the way to the destination.



• The routers are too slow to perform bookkeeping tasks (queuing buffers, updating tables, etc.).
• The routers' buffer is too limited.
• Congestion in a subnet can occur if the processors are slow. Slow speed CPU at routers will perform the routine tasks such as queuing buffers, updating table etc slowly. As a result of this, queues are built up even though there is excess line capacity.

- Congestion is also caused by slow links. This problem will be solved when high speed links are used. But it is not always the case. Sometimes increase in link bandwidth can further deteriorate the congestion problem as higher speed links may make the network more unbalanced.Congestion can make itself worse. If a route!” does not have free buffers, it start ignoring/discarding the newly arriving packets.
- When these packets are discarded, the sender may retransmit them after the timer goes off. Such packets are transmitted by the sender again and again until the source gets the acknowledgement of these packets. Therefore multiple transmissions of packets will force the congestion to take place at the sending end.

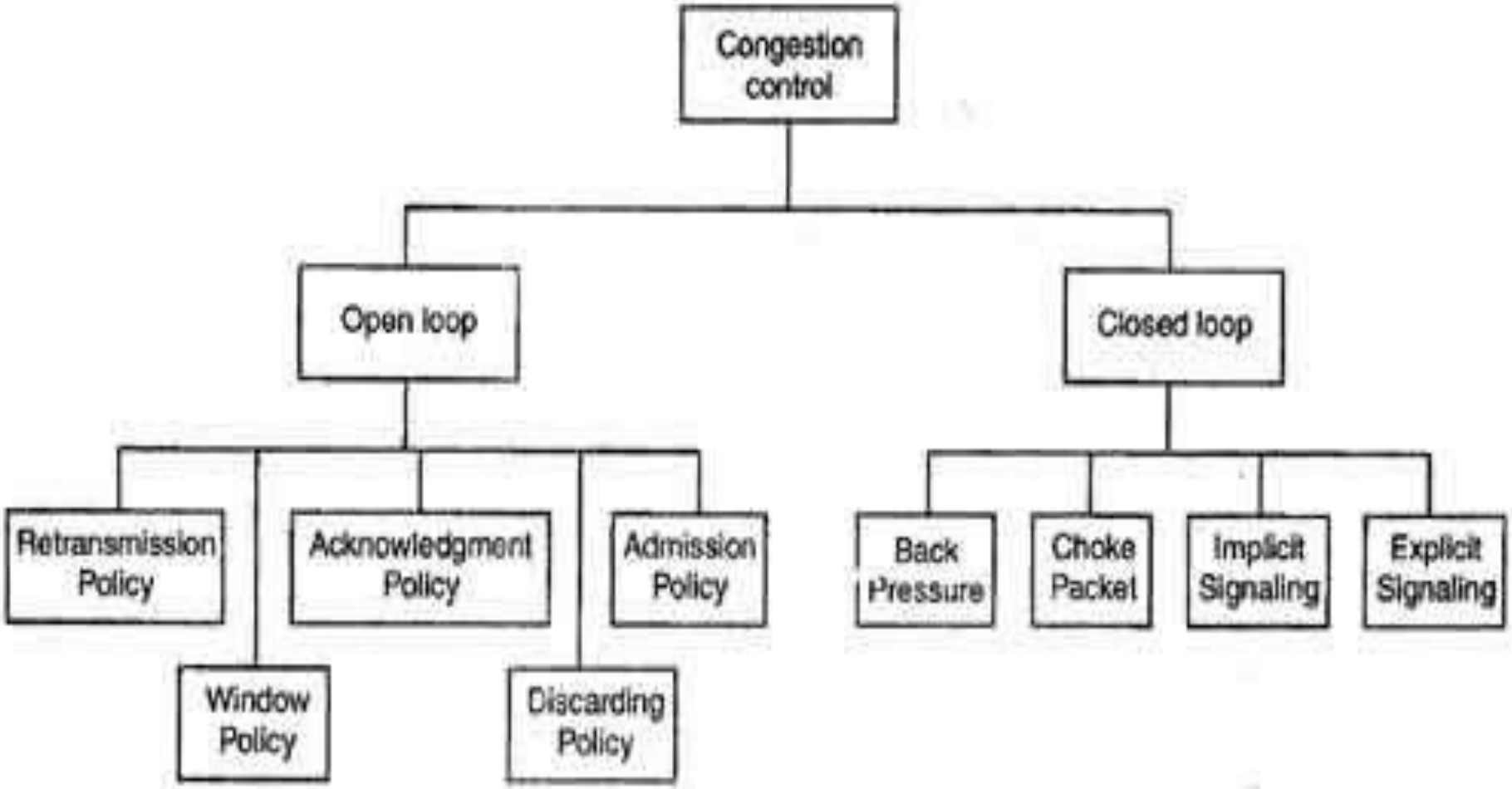
difference between congestion control and flow control.

- Congestion control has to do with making sure the network is able to carry the offered traffic. It is a global issue, involving the behavior of all the hosts and routers.
- Flow control, in contrast, relates to the traffic between a particular sender and a particular receiver. Its job is to make sure that a fast sender cannot continually transmit data faster than the receiver is able to absorb it.
- To see the difference between these two concepts, consider a network made up of 100-Gbps fiber optic links on which a supercomputer is trying to force feed a large file to a personal computer that is capable of handling only 1 Gbps. Although there is no congestion (the network itself is not in trouble), flow control is needed to force the supercomputer to stop frequently to give the personal computer a chance to breathe.

- At the other extreme, consider a network with 1-Mbps lines and 1000 large computers, half of which are trying to transfer files at 100 kbps to the other half. Here, the problem is not that of fast senders overpowering slow receivers, but that the total offered traffic exceeds what the network can handle.
- The reason congestion control and flow control are often confused is that the best way to handle both problems is to get the host to slow down. Thus, a host can get a ‘‘slow down’’ message either because the receiver cannot handle the load or because the network cannot handle it.

How to correct the Congestion Problem:

- Congestion Control refers to techniques and mechanisms that **can either prevent congestion, before it happens,** or **remove congestion, after it has happened.**
- Congestion control mechanisms are divided into two categories, **one category prevents the congestion from happening(policies are used to prevent the congestion before it happens).**
- And the **other category removes congestion after it has taken place(policies are used to remove the congestion after it happens.).**



Types of Congestion Control Methods

Congestion control prevention policies:

These two categories are:

1. Open loop
2. Closed loop

1. Open Loop Congestion Control

- **Policies adopted by open loop congestion control**
- In this method, **policies are used to prevent the congestion before it happens.**
- Congestion control is handled either by the source or by the destination.
- The various methods used for open loop congestion control are:

A. Retransmission Policy

- The sender can retransmits a packet, if packet is lost or corrupted.
- However retransmission in general may increase the congestion in the network. But we need to implement good retransmission policy to prevent congestion.

B. Window Policy

- To implement window policy, selective repeat window method is used for congestion control.
- Selective repeat method sends only the specific lost or damaged packets.

C. Acknowledgement Policy

- The acknowledgement policy imposed by the receiver may also affect congestion.

• If the receiver does not acknowledge every packet it receives it may slow down the sender and help prevent congestion.

• Acknowledgments also add to the traffic load on the network. Thus, by sending fewer acknowledgements we can reduce load on the network.

• To implement it, several approaches can be used:

1. A receiver may send an acknowledgement only if it has a packet to be sent.

2. A receiver may send an acknowledgement when a timer expires.

D. Discarding Policy

• A router may discard less sensitive packets when congestion is likely to happen.

• Such a discarding policy may prevent congestion and at the same time may not harm the integrity of the transmission.

E. Admission Policy

• **An admission policy, which is a quality-of-service mechanism, can also prevent congestion in virtual circuit networks.**

• Switches in a flow first check the resource requirement of a flow before admitting it to the network.

• **A router can deny establishing a virtual circuit connection** if there is congestion in the "network or if there is a possibility of future congestion.

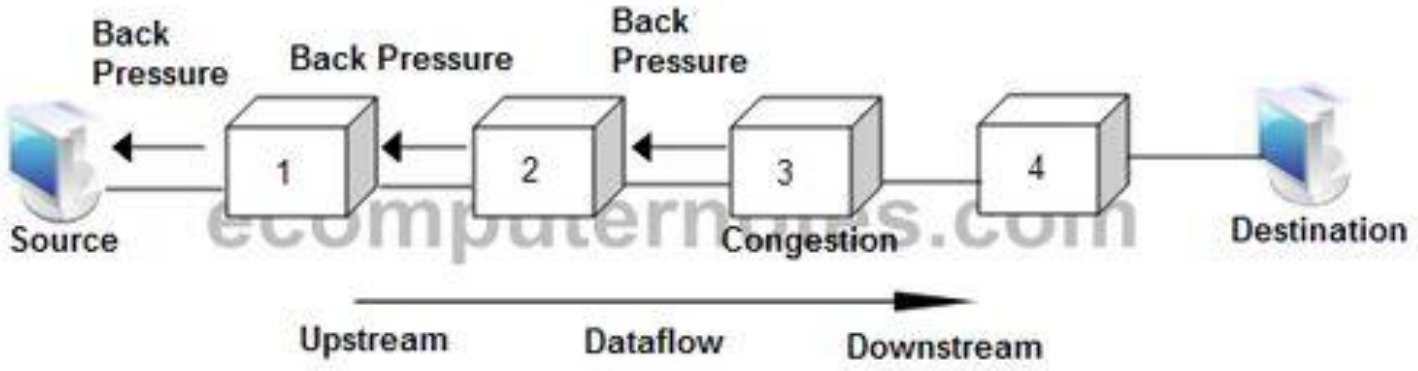
2. Closed Loop Congestion Control

• **Closed loop congestion control mechanisms try to remove the congestion after it happens**.

• The various methods used for closed loop congestion control are:

A. Backpressure

• Back pressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow.



Backpressure Method

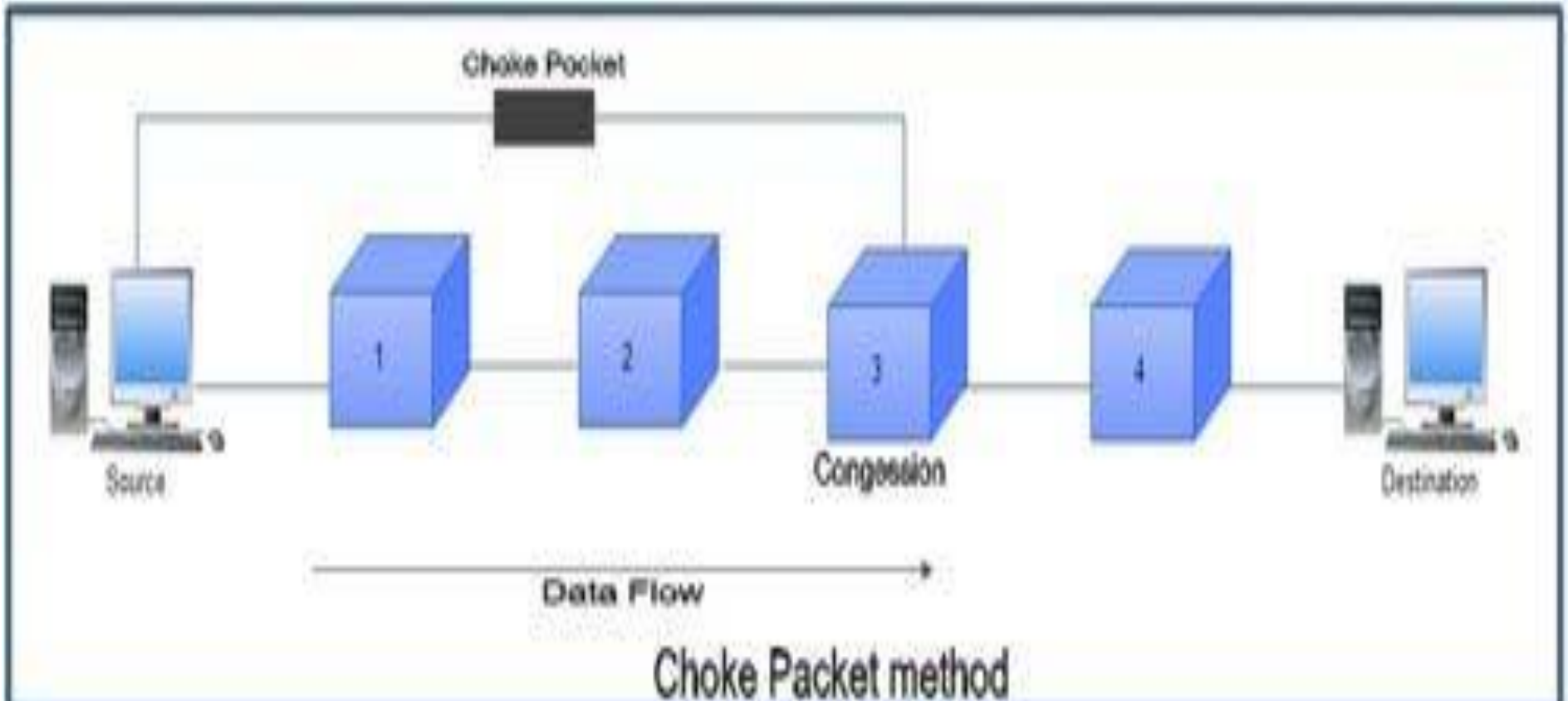
- The backpressure technique can be applied only to virtual circuit networks. In such virtual circuit each node knows the upstream node from which a data flow is coming.
- In this method of congestion control, the congested node stops receiving data from the immediate upstream node or nodes.
  - This may cause the upstream node on nodes to become congested, and they, in turn, reject data from their upstream node or nodes.

• As shown in fig node 3 is congested and it stops receiving packets and informs its upstream node 2 to slow down. Node 2 in turns may be congested and informs node 1 to slow down. Now node 1 may create congestion and informs the source node to slow down. In this way the congestion is alleviated. Thus, the pressure on node 3 is moved backward to the source to remove the congestion.

B. Choke Packet

• In this method of congestion control, congested router or node sends a special type of packet called choke packet to the source to inform it about the congestion.

• Here, congested node does not inform its upstream node about the congestion as in backpressure method.



Choke Packet method

- In choke packet method, congested node sends a warning directly to the source station *i.e.* the intermediate nodes through which the packet has traveled are not warned.

D. Implicit Signaling

- In implicit signaling, there is no communication between the congested node or nodes and the source.
- The source guesses that there is congestion somewhere in the network when it does not receive any acknowledgment. Therefore the delay in receiving an acknowledgment is interpreted as congestion in the network.
- On sensing this congestion, the source slows down.
- This type of congestion control policy is used by TCP.

E. Explicit Signaling

- In this method, **the congested nodes explicitly send a signal to the source or destination to inform about the congestion.**
- Explicit signaling is different from the choke packet method. In choke packed method, a separate packet is used for this purpose whereas in explicit signaling method, the signal is included in the packets that carry data .
- Explicit signaling can occur in either the forward direction or the backward direction .

- In backward signaling, a bit is set in a packet moving in the direction opposite to the congestion. This bit warns the source about the congestion and informs the source to slow down.
- In forward signaling, a bit is set in a packet moving in the direction of congestion. This bit warns the destination about the congestion. The receiver in this case uses policies such as slowing down the acknowledgements to remove the congestion.

Congestion control algorithms

Traffic shaping

This method of congestion control is to "shape" the traffic before it enters the network.

- In computer networks, one of the main reasons for congestion is that data is not sent at constant rates but rather in bursts called traffic peaks. Handling these traffic peaks is quite simple. Controlling the sending rate on network participants must flatten traffic peaks. **This kind of traffic control is a preventive Congestion Avoidance mechanism called Traffic Shaping.**

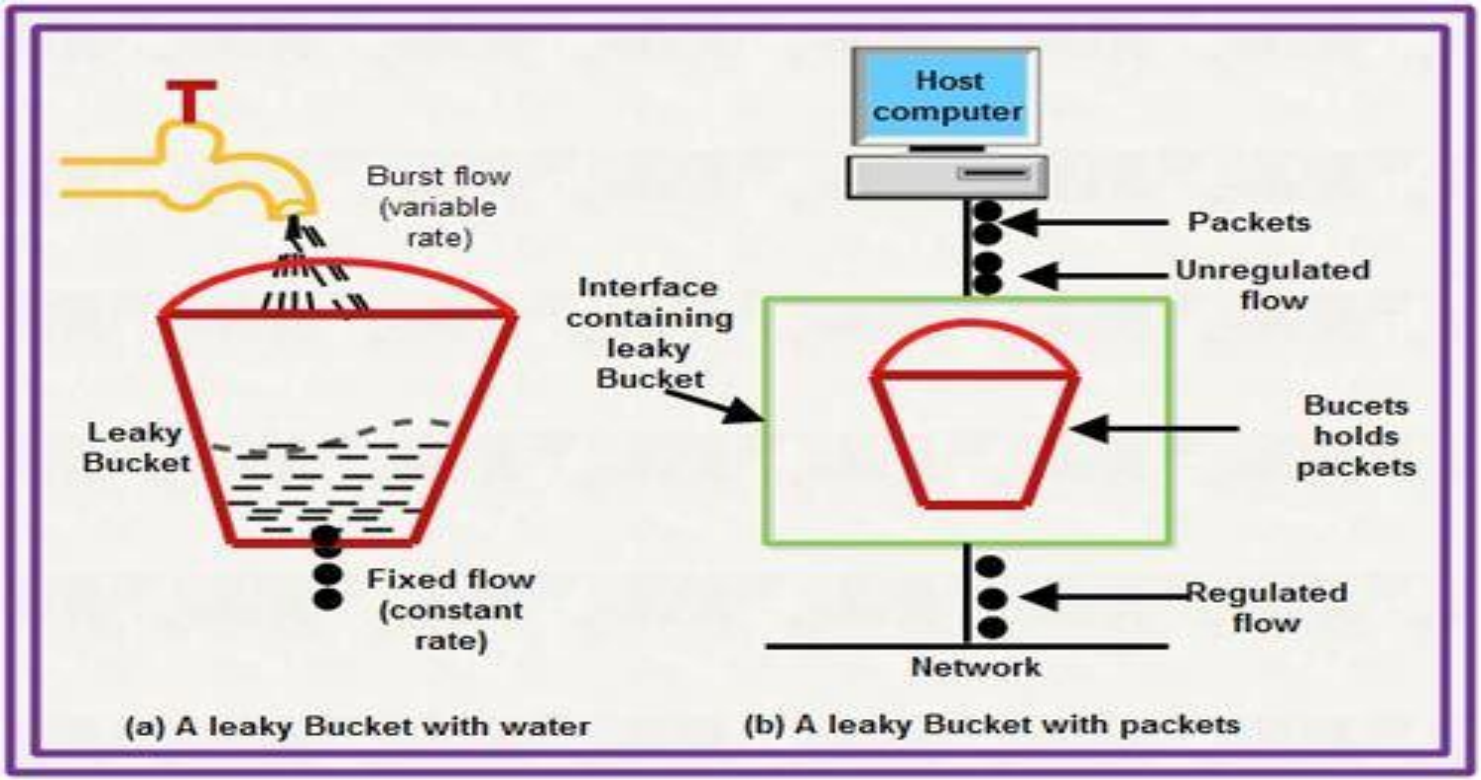
Two techniques can shape traffic:

1. Leaky Bucket Algorithm

It is a traffic shaping mechanism that controls the amount and the rate of the traffic sent to the network.

• A leaky bucket algorithm shapes bursty traffic into fixed rate traffic by averaging the data rate.

- Imagine a bucket with a small hole at the bottom.
- The rate at which the water is poured into the bucket is not fixed and can vary but it leaks from the bucket at a constant rate. Thus (as long as water is present in bucket), the rate at which the water leaks does not depend on the rate at which the water is input to the bucket.



- Also, when the bucket is full, any additional water that enters into the bucket spills over the sides and is lost.
- The same concept can be applied to packets in the network. Consider that data is coming from the source at variable speeds. Suppose that a source sends data at 12 Mbps for 4 seconds. Then there is no data for 3 seconds. The source again transmits data at a rate of 10 Mbps for 2 seconds. Thus, in a time span of 9 seconds, 68 Mb data has been transmitted.
- If a leaky bucket algorithm is used, the data flow will be 8 Mbps for 9 seconds. Thus constant flow is maintained.

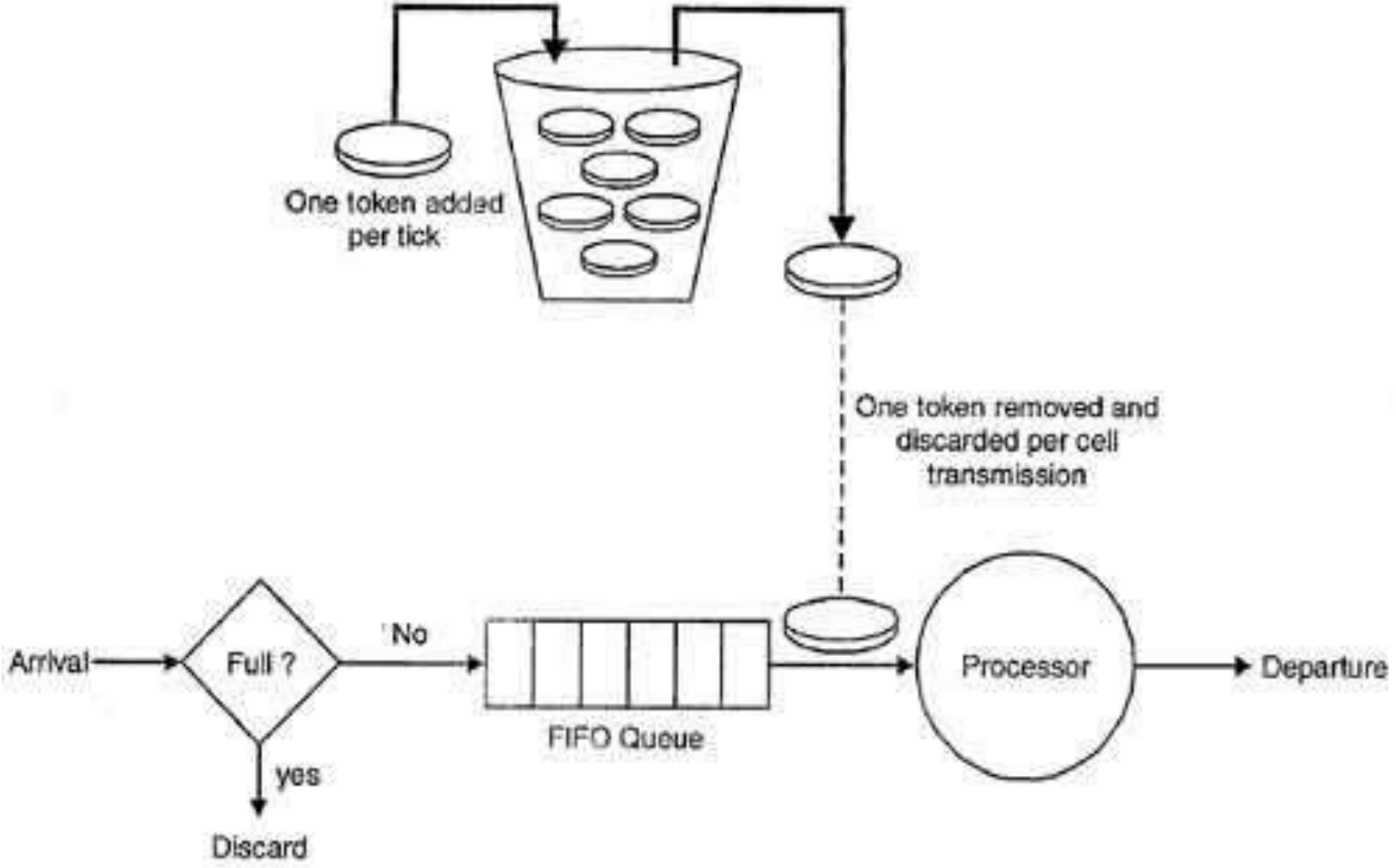
Each network interface contains a leaky bucket and the **following steps are involved in leaky bucket algorithm:**

- When host wants to send packet, packet is thrown into the bucket.
- The bucket leaks at a constant rate, meaning the network interface transmits packets at a constant rate.
- Bursty traffic is converted to a uniform traffic by the leaky bucket.
- In practice the bucket is a finite queue that outputs at a finite rate.

2. Token bucket Algorithm

- The leaky bucket algorithm allows only an average (constant) rate of data flow. Its major problem is that it cannot deal with bursty data.
- A leaky bucket algorithm does not consider the idle time of the host. For example, if the host was idle for 10 seconds and now it is willing to sent data at a very high speed for another 10 seconds, the total data transmission will be divided into 20 seconds and average data rate will be maintained. The host is having no advantage of sitting idle for 10 seconds.

- To overcome this problem, a token bucket algorithm is used. A token bucket algorithm allows bursty data transfers.
- A token bucket algorithm is a modification of leaky bucket in which leaky bucket contains tokens.
- In this algorithm, a token(s) are generated at every clock tick. For a packet to be transmitted, system must remove token(s) from the bucket.
- Thus, a token bucket algorithm allows idle hosts to accumulate credit for the future in form of tokens.
- For example, if a system generates 100 tokens in one clock tick and the host is idle for 100 ticks. The bucket will contain 10,000 tokens.
- Now, if the host wants to send bursty data, it can consume all 10,000 tokens at once for sending 10,000 cells or bytes.
- Thus a host can send bursty data as long as bucket is not empty.



Token bucket algorithm

Steps of this algorithm can be described as follows:

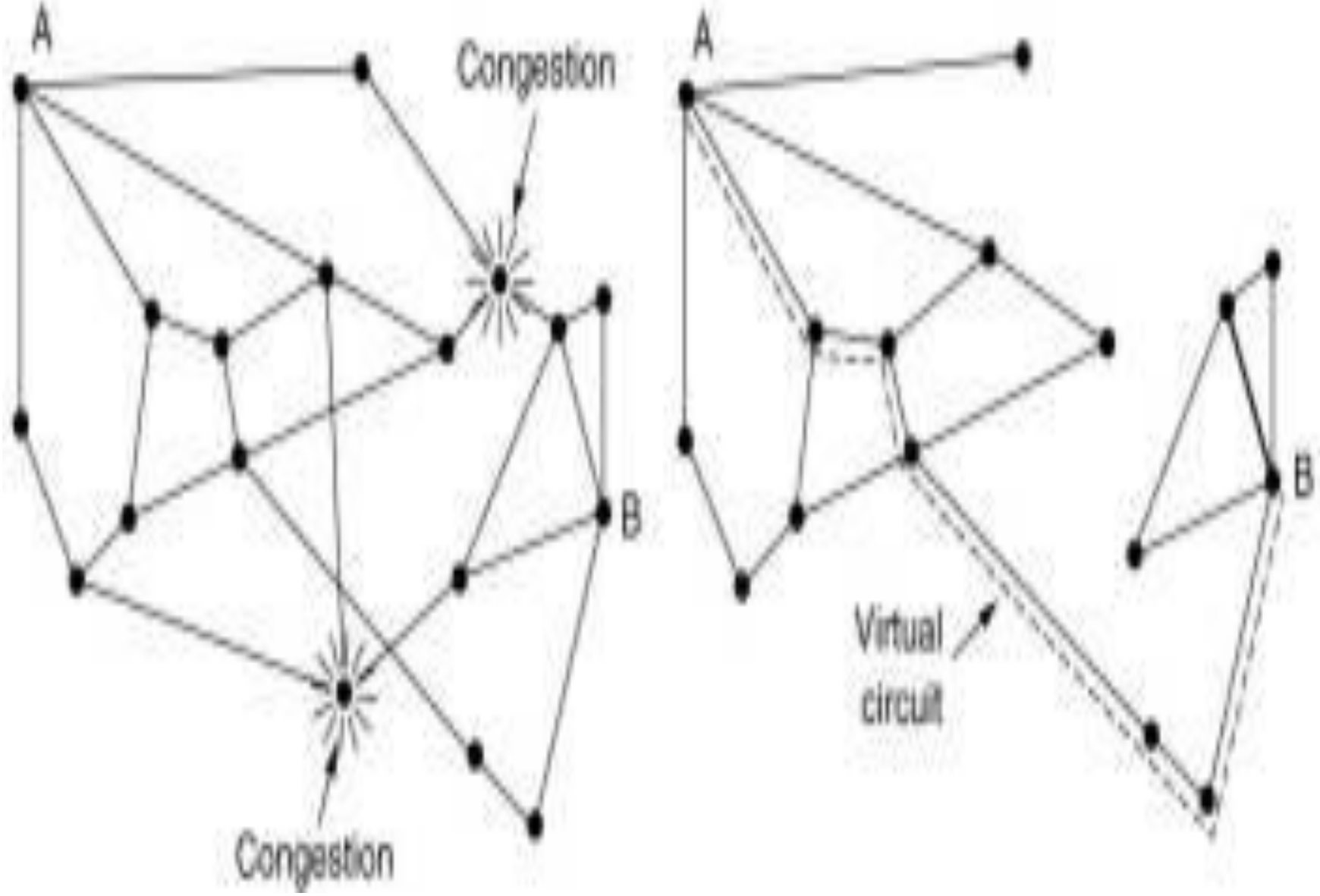
- In regular intervals tokens are thrown into the bucket. *f*
- The bucket has a maximum capacity. *f*
- If there is a ready packet, a token is removed from the bucket, and the packet is sent.
- If there is no token in the bucket, the packet cannot be sent.

Leaky Bucket vs. Token Bucket

- LB discards packets; TB does not. TB discards tokens.
- With TB, a packet can only be transmitted if there are enough tokens to cover its length in bytes.
- LB sends packets at an average rate. TB allows for large bursts to be sent faster by speeding up the output.
- TB allows saving up tokens (permissions) to send large bursts. LB does not allow saving.

Congestion control for virtual circuit subnets

- With **admission control** once congestion has been signaled, no more virtual circuits are set up until the problem has gone away. This is a crude approach, but it is simple and easy to implement.
- Another way is to route new connections around the problem area. The subnet is "redrawn" leaving congested routers and all their lines out and then determine the best route for a new connection in that subnet. A step further is to try to avoid routers that are directly connected to the congested routers.

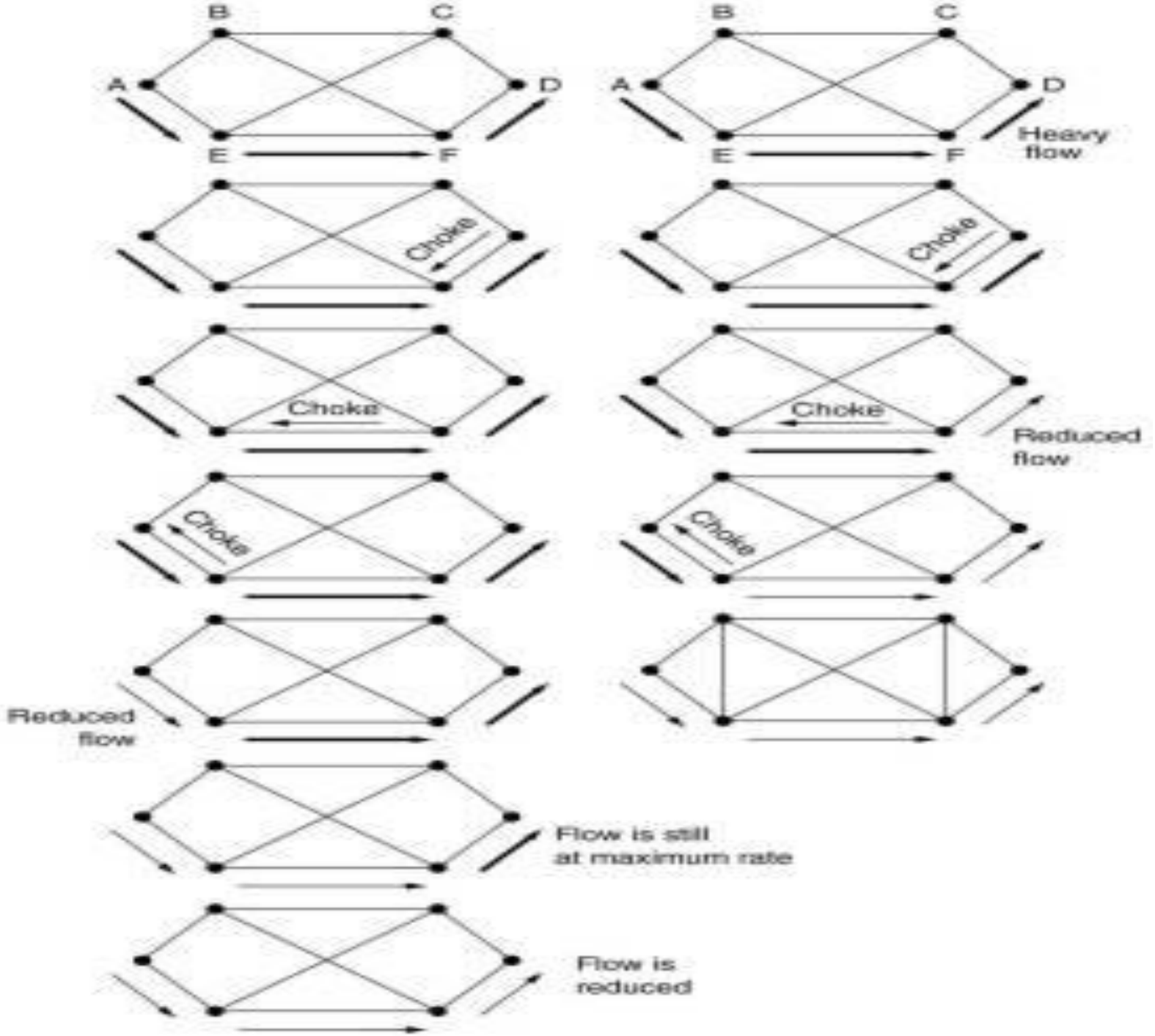

A
Congestion
B
Congestion
A
Virtual
circuit
B

- Another strategy is to negotiate an agreement between the host and subnet when the connection is set up. This would specify the volume and shape of the traffic, quality of service and other parameters. To keep its part of the agreement, the subnet will typically reserve resource along the path the VC is set up.
- This can include table and buffer space in the routers and bandwidth on the lines. In this way congestion can be avoided. This kind of reservation can be done always, but this tends to waste resources.
- If 6 VC's that might use 1 Mbps pass through a 6 Mbps line, this line has to be considered full, even though it may rarely happen that all 6 VC are transmitting at full speed at the same time. The reservation can also only be done when there is a congestion on the subnet.
- In general the price of congestion control is unused and wasted bandwidth.

Congestion control for datagram subnets

- Each router can easily monitor the utilization of its output lines and other resources. **In the old DECNET a warning bit** was set in the header of a packet send out over a too busy line or when the router is low on buffer space. At the destination, the transport entity copied this bit into the next acknowledgment sent to the source, which reacts by reducing the transmission rate. This continues when more warning bits come in. When no more warning bits arrive, the transmission rate is increased slowly.

- A router can also directly send a **choke packet** back to the source host of a packet send out over a too busy line. That packet is tagged so that it will not generate more choke packets on its way. When the source host receives the choke packet, it reduces its rate to the specified destination by a certain fraction. If it receives a choke packet in reaction to its new rate, it reduces again by that fraction.
- For high speeds or long distances, the reaction is too slow. With **hop-by-hop choke packets** each intermediate router also reacts on a choke packet by reducing its sending rate. For that it needs sufficient buffers to store the packets which still come in at a too high rate.


B
C
A
D
E
F
Heavy flow
Choke
Choke
Choke
Reduced flow
Choke
Choke
Reduced flow
Flow is still at maximum rate
Flow is reduced

Load Shedding

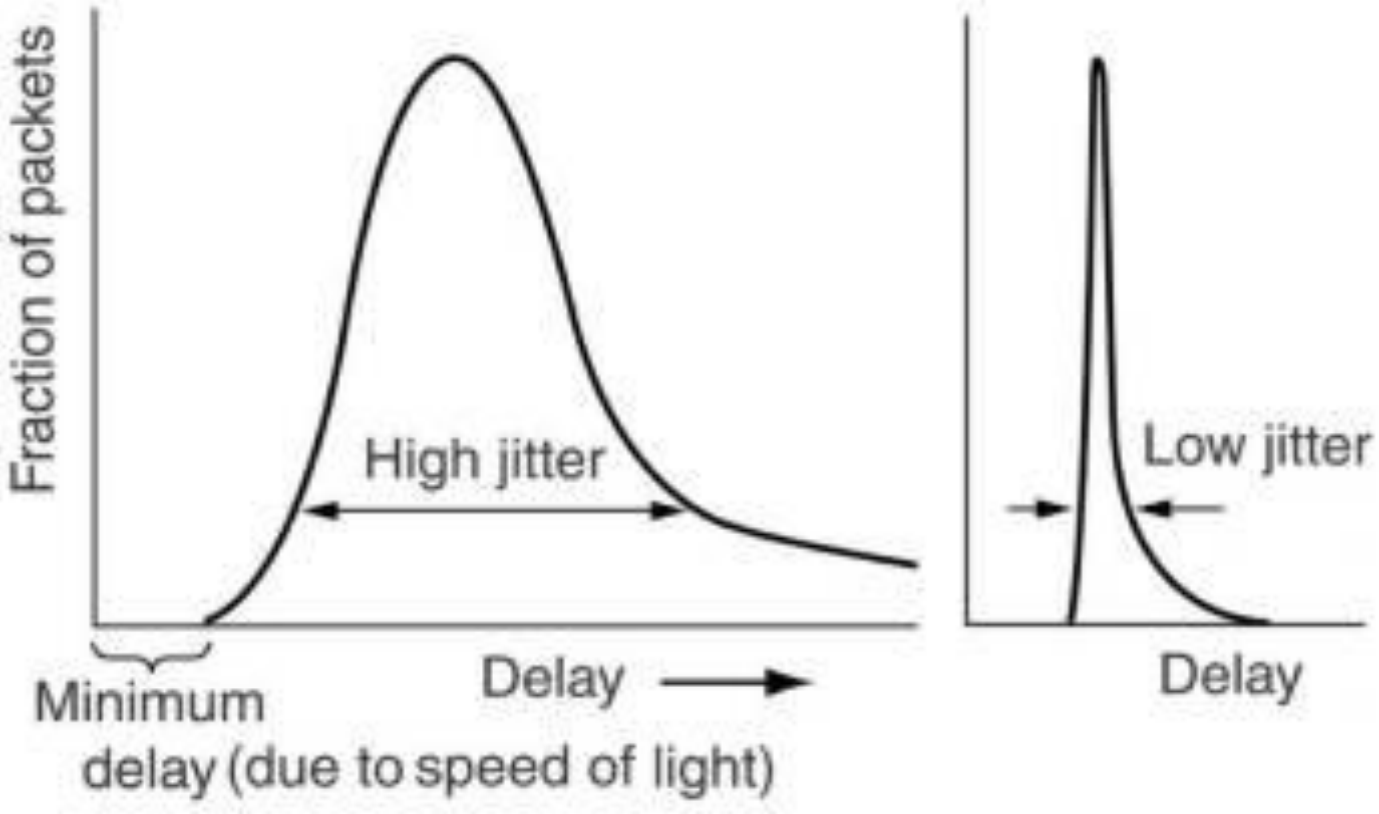
- A fancy way of saying that when routers are being inundated by packets that they cannot handle, they just throw them away. Depending on the application the packet is used for, this can be done so that triggered retransmissions are reduced. For example, for file transfers it is best to throw away newer packets, for multimedia older packets are best.
- Applications can indicate the priority and action in case of congestion of their packets in the headers of them, for example in IP.
- In **random early detection** routers drop packets before the situation has become hopeless, for example when their buffers are filled above a certain fraction. As it is difficult to tell which source contributes most to the trouble, a random selection of the packets to drop is used.

- A choke packet could be sent to the source of a dropped packet, with a disadvantage that it puts even more load on an already congested network. Fortunately, in most transport protocols designed for rather reliable wired networks, like TCP, it was realized that lost packets are mostly due to buffer overruns somewhere rather than transmission errors. These protocols respond to a lost packet by temporarily reducing the sending rate, so a choke packet is not needed there. In wireless networks most losses are due to noise on the air link, here choke packets are needed.

Jitter control :

- For applications such as audio and video streaming, it does not matter much if the packets take 20 or 30 msec to be delivered, as long as the transit time is constant. The standard deviation in transit times is called **jitter**
- This can be bounded by computing the expected transit time for each hop. When a packet arrives at a router, it is checked how much the packet is behind or ahead its schedule. This is stored in the header and updated at each hop. The router sends then always the packet which is furthest behind.

- In some applications, like video on demand, jitter can be compensated for by buffering at the receiver. For others, like Internet telephony or videoconferencing, the delay inherent in buffering is not acceptable.



Quality of Service(QOS)

- **QOS is an interworking issue which can be defind as flow which is required for communication.**
- **QoS** is determined by **four primary parameters: reliability, delay, jitter and bandwidth.** The table shows how stringent these requirements are for various applications.

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

The needs of each flow can be characterized by four primary parameters :

- **Reliability:** Lack of reliability means losing a packet or acknowledgement which entertains retransmission.
- **Delay:** Increase in delay means destination will find the packet later than expected, Importance of delay changes according to the various application.
- **Jitter:** Variation of the delay is jitter, If the delay is not at a constant rate, it may result in poor quality.
- **Bandwidth:** Increase in bandwidth means increase in the amount of data which can be transferred in given amount of time, Importance of bandwidth also varies according to various application.

Techniques for achieving good Quality of Service :

Overprovisioning –

1. Overprovisioning means to provide so much router capacity, buffer space and bandwidth that all the packets just fly through easily. The problem with this is that it is expensive. But as time goes on and design and technology improves, it sometimes even becomes practical. The telephone system is an example, it is rare to pick up a telephone and not get a dial tone instantly.

2. Buffering –

Flows can be buffered on the receiving size before being delivered to a monitor or speaker. It does not affect the reliability or bandwidth, it increases the delay, but it smoothes out the jitter.

- For audio and video on demand, jitter is a main problem, so it helps here. Commercial web sites that contain streaming audio or video all use players that buffer for about 10 seconds before starting to play. If the bandwidth is low, much more must be buffered, or the whole play must be first stored on a local disk.

3. Traffic Shaping –

- It is defined as about regulating the average rate of data transmission.
- It smooths the traffic on server side other than client side.
- When a connection is set up, the user machine and subnet agree on a certain traffic pattern for that circuit called as Service Level Agreement. It reduces congestion and thus helps the carrier to deliver the packets in the agreed pattern.
- The **leaky bucket algorithm** to regulate the maximum number of packets (or bytes) per time unit. When the bucket is full, packets are discarded.

- When the packets are all the same size (as in ATM), the algorithm can be used as described. When variable-sized packets are used it is often better to allow for a fixed number of bytes per tick.
- The **token bucket algorithm** allows the output rate to speed up to a certain maximum when a large burst arrives from the host computer. Tokens are used to accumulate rights to send packets or bytes. Packets are less likely to be discarded.
- Sometimes a token bucket is followed by a leaky bucket.

Example, ATM(Asynchronous Transfer Mode) networks classify flows into 4 broad categories with respect to their QoS demands, a division which is also useful for other networks:

1. Constant bit rate (e.g. telephony), attempts to simulate a wire, providing uniform bandwidth and delay.
2. Variable bit rate (e.g. compressed videoconferencing),images must arrive in time independent on how much they could be compressed.
3. Non-real-time variable bit rate (e.g. watching a movie over internet), a lot of buffering at the receiver is allowed.
4. Available bit rate (e.g. file transfer), applications that are not sensitive to jitter or delay.

Internetworking

- The word “internetworking,” which combines the words “inter” and “networking,” denotes a connection between completely distinct nodes/segments. This connection is made possible by intermediary hardware like routers or gateways.
- In internetworking, routers are aware of each other’s addresses as well as addresses outside of their network. They have two options: they can be statically set to operate on various networks, or they can learn by utilising the internetworking routing protocol.

How Does Internetworking Work?

- Every network node or phase is built using a similar protocol or a communication logic, such as TCP (Transfer Control Protocol) or IP (Internet Protocol), to enable communication. It is referred to as “internetworking” when a network interacts with another network using ongoing communication protocols. A packet of information must be delivered across numerous links, which is a problem that internetworking was created to address.
- The distinction between expanding the network and internetworking is quite slight. A simple extension of a LAN is the use of a switch or hub to join two local area networks, but connecting them via a router is an example of internetworking. The OSI-ISO model’s Layer 3 (Network Layer) enforces internetworking. The internet is the most prominent famous example of internetworking.

Types of Internetworking

- **Internetworking primarily consists of three units: Extranet, Internet, and Intranet.**
- Internetworking primarily consists of three units: Extranet, Internet, and Intranet. Internet connections may or may not be present on intranets and extranets. The computer network or the extranet area unit is typically protected from being accessed from the internet if it is not approved and if there is a link to the internet. Although it should serve as a portal for access to portions of the associate degree extranet, the internet is not considered to be a part of the computer network or extranet.

Extranet

- It's a network of the internetwork with a confined scope to one organisation or institution but with limited links to one or more other networks on occasion; however, this is not always the case.
- It is the lowest degree of internet usage and is typically prohibited in extremely private areas. An extranet may also be referred to as a MAN, WAN, or another type of network, but it cannot include a single local area network; rather, it must make at least one mention of an external network.

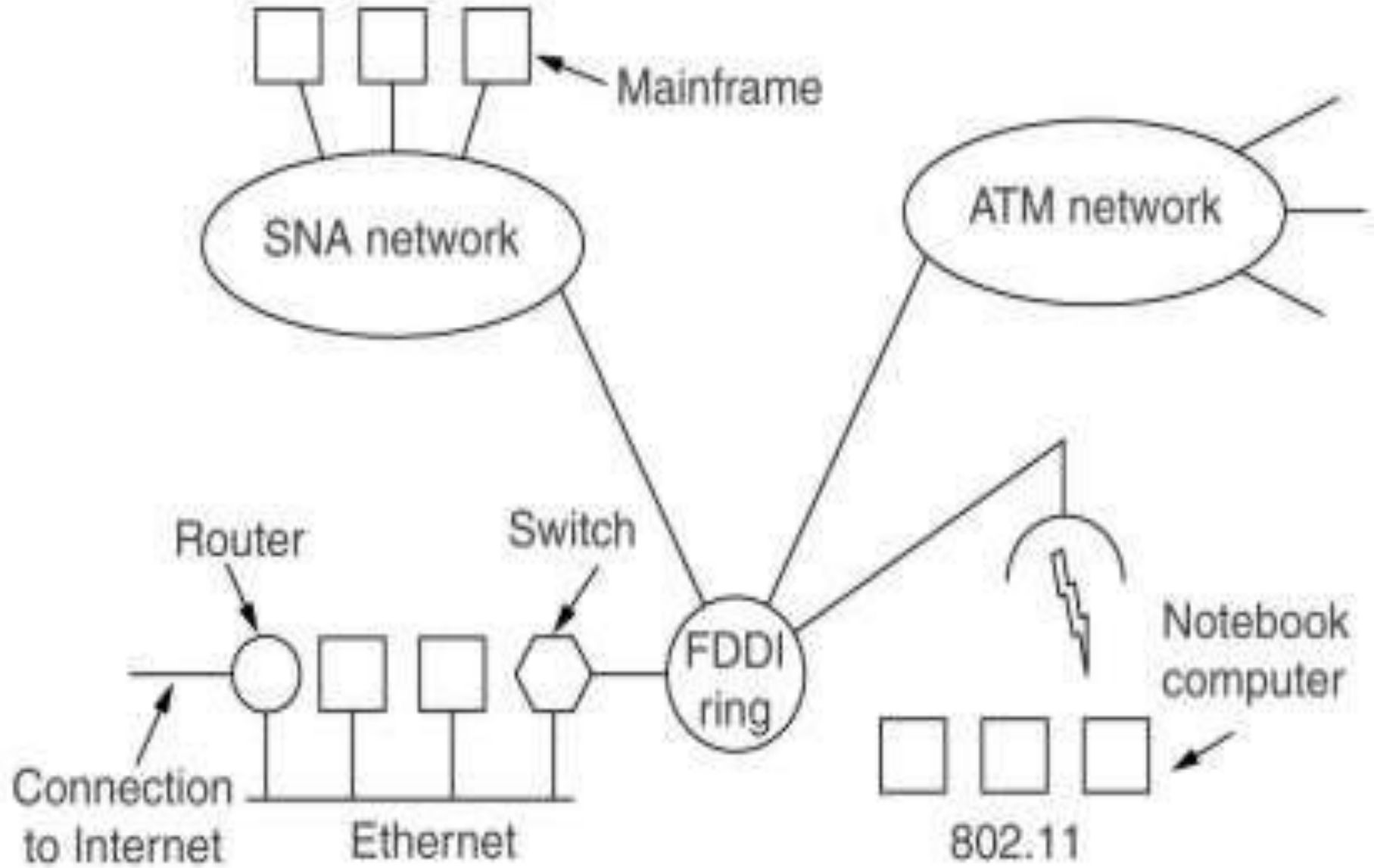
Internet

- Internet is a specific internetworking that connects governmental, academic, public, and private networks on a global scale. It is based on the ARPANET, which was created by the ARPA (Advanced Research Projects Agency) of the U.S. Defense Department. It is also the location of the World Wide Web (WWW) and is referred to as the "Internet" to distinguish it from other generic internetworking. Internet users and their service providers utilise IP addresses obtained from address registries that control assignments.

Intranet

- This computer network can be a collection of interconnected networks that employ the Internet Protocol and IP-based software like web browsers as well as FTP tools, all of which are controlled by a single body entity. This body entity blocks access to the computer network for the rest of the world and only allows a select few users.
- This network most frequently refers to the internal network of a business or other enterprise. To provide users with browseable data, a large computer network can typically have its own internet server.

- Various boxes working on different protocol layers can be used for internetworking.



Construct some gateways that connect different kinds of networks –

- Repeaters or hubs at physical layer: operate on bits, do not understand protocols, just regenerate signals
- Bridges or switches at data link layer: operate on frames, examine MAC addresses, do minor protocol translation, e.g. Ethernet to 802.11
- Multiprotocol router at network layer: operate on packets, translate between different packet formats, possibly split up packets
- Transport gateways at transport layer: coupling byte streams in different networks, interface between different transport connections
- Application gateways at application layer: translate message semantics between OSI and TCP/IP networks

How networks differ

Differences that can occur at network layer, which makes internetworking difficult:

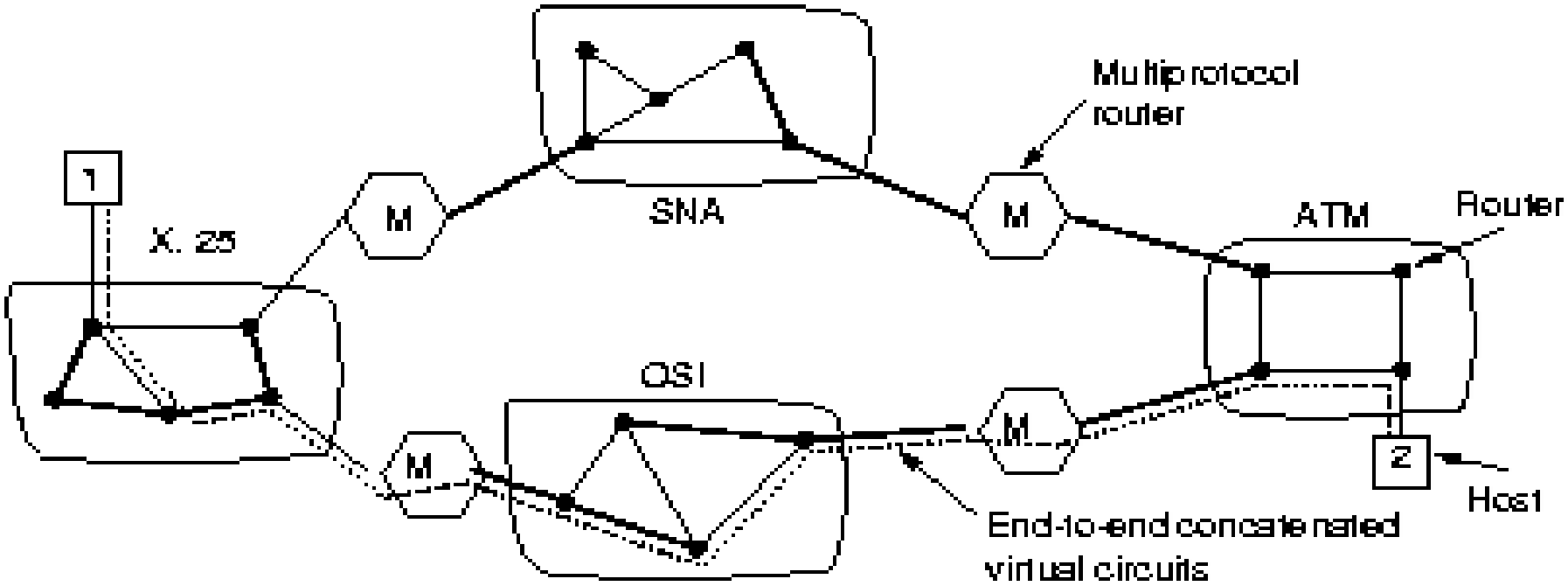
Item	Some Possibilities
Service offered	Connection oriented versus connectionless
Protocols	IP, IPX, SNA, ATM, MPLS, AppleTalk, etc.
Addressing	Flat (802) versus hierarchical (IP)
Multicasting	Present or absent (also broadcasting)
Packet size	Every network has its own maximum
Quality of service	Present or absent; many different kinds
Error handling	Reliable, ordered, and unordered delivery
Flow control	Sliding window, rate control, other, or none
Congestion control	Leaky bucket, token bucket, RED, choke packets, etc.
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, by packet, by byte, or not at all

How networks can be connected

- With a router, packet is extracted from frame and address in packet is used to decide where to send it → so router has to understand network protocols
- In Physical layer, networks can be connected by repeater or hubs, which just move the bits from one network to an identical network.
- In data link layer, network can be connected by switch and bridge. They can accept frames, examine the MAC address, and forward the frames to a different network.
- In the network (IP) layer, we have routers that can connect two networks. In the transport and application we have gateways.

Concatenating virtual circuits

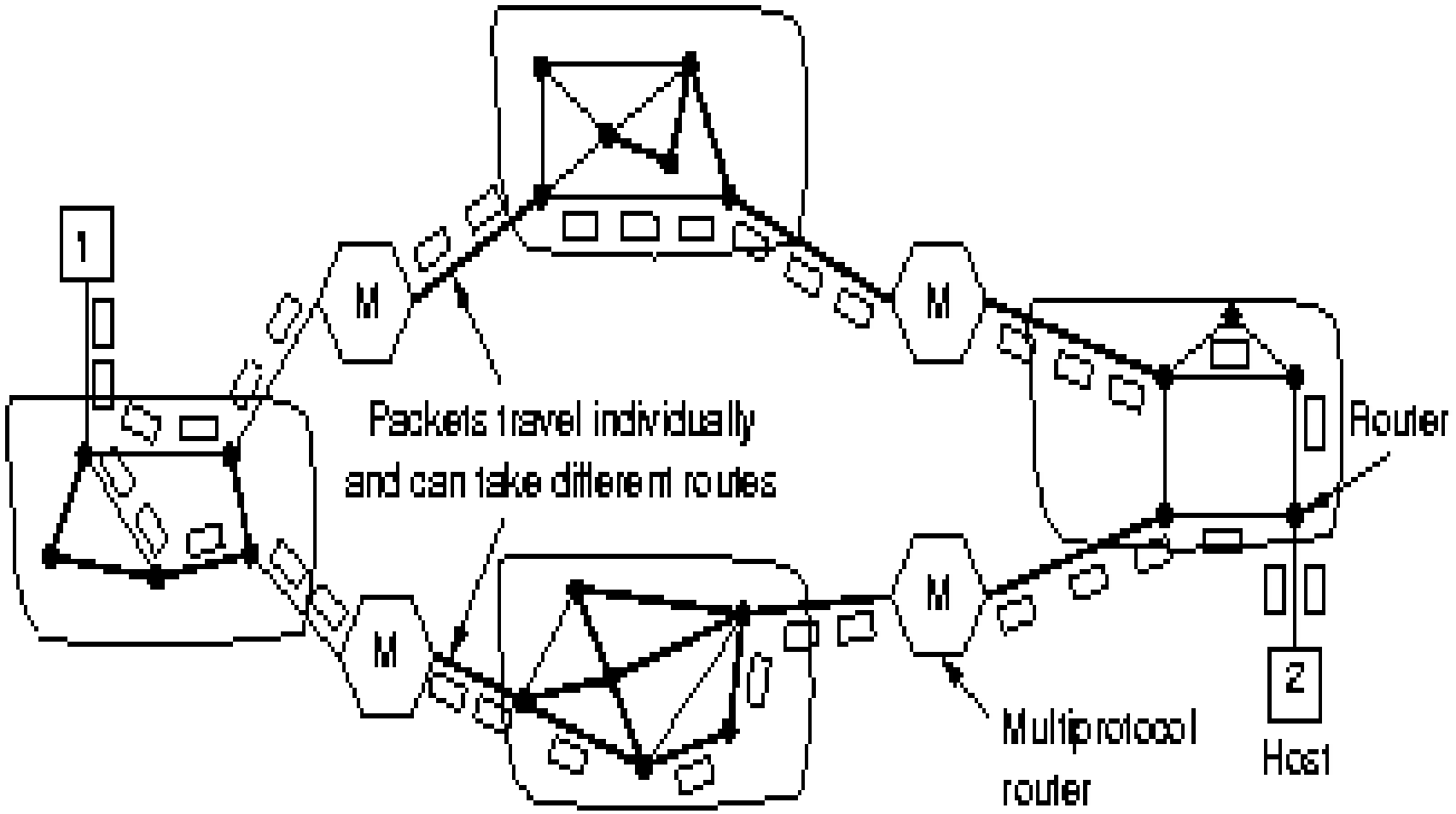
- A sequence of virtual circuits is set up from source through gateways (multiprotocol routers) to destination
- A sequence of virtual circuits is set up from the source trough one or more gateways to the destination. Each gateway maintains tables telling which virtual circuits pass through it, where they are to be routed, and what the new virtual circuit number is.



- Once data packets begin flowing, each gateway relays incoming packets, converting between packet formats and VC numbers as needed. As all packets must transverse the same sequence of gateways, they arrive in order.
- This scheme works best when all the networks have roughly the same properties. It can not be used if one of the subnets does not offer VC's but only datagrams.

Connectionless internetworking

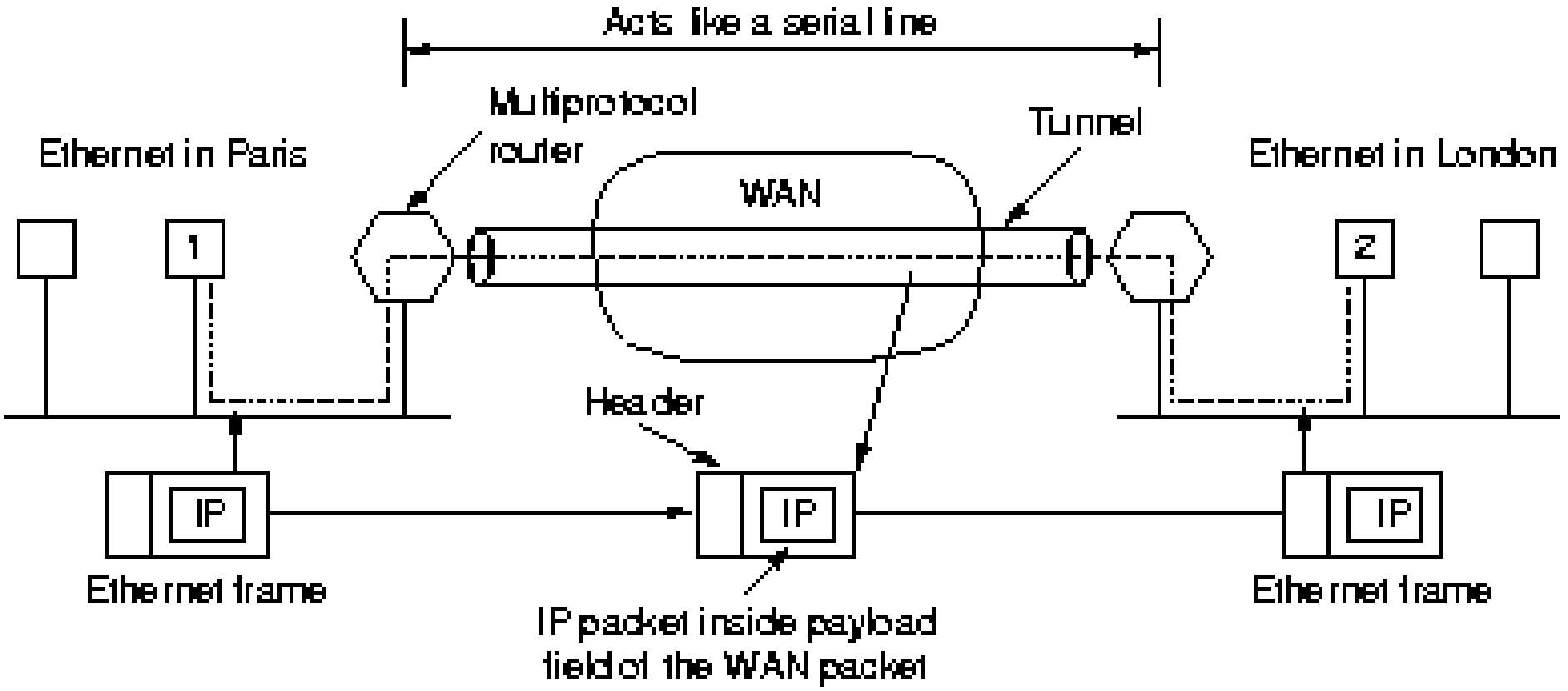
- This connectionless internetworking approach is similar to datagram service in subnets .



- As a routing decision is made for each packet, possible depending on the traffic, packets might arrive out of order. But a higher bandwidth might be achieved than when concatenating VC's.
- Addressing is a problem if e.g. an IP packet with 32 bit addresses has to be send to an OSI host with decimal addresses similar to telephone numbers. One would need an IP number for the OSI host and an OSI number for the IP host, to make the conversion in the multiprotocol routers. This is impossible on a large scale with the current IPv4 protocol.
- The properties of the datagram approach to internetworking are the same as those of datagram subnets: more potential for congestion but also more potential for adapting to it, robustness in the face of router failures and longer packet headers.

Tunneling

- Solving general internetworking problem is extremely hard, but thing becomes easier if source and destination are on same type of subnet – tunnelling can be used .

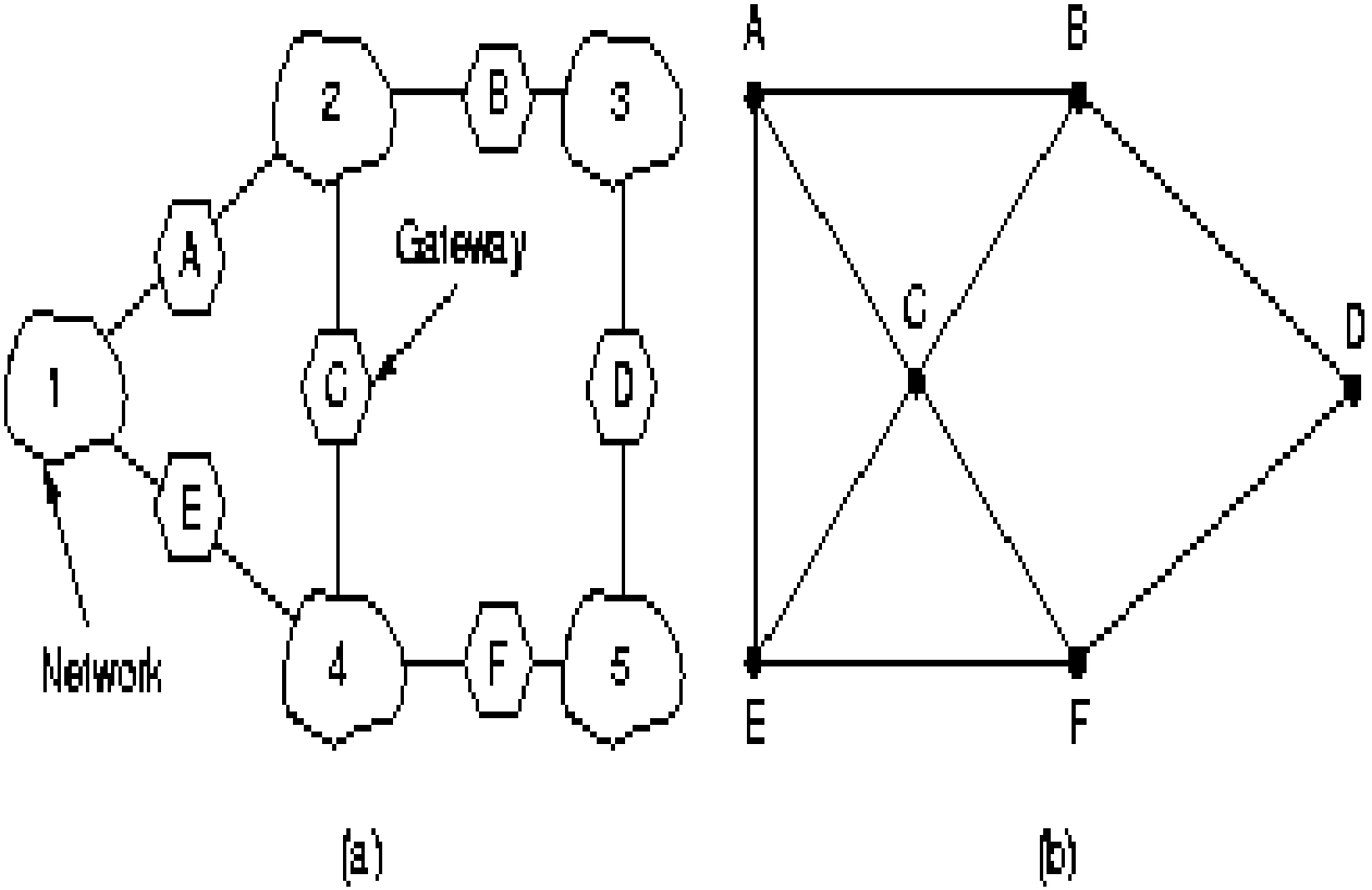


- As an example, think of an international bank with an IPv6 network in Paris, an IPv6 network in London and connectivity between the offices via the IPv4 Internet.
- Multiprotocol router simply places source host's packet inside payload of WAN packet and sends through.
- Essentially, what between two hosts' subnets, WAN, can be seen as a "tunnel" extending from one multiprotocol router to the other.

Internetwork routing

- This is similar to routing within a single subnet with some additional complications.
- **Routers might** prefer routes with no protocol conversions over ones that use tunneling over ones that need protocol conversions, even if those routes are longer.
- **Subnets might** be run by different carriers and have different charging algorithm, so routing might be based on cost.
- When international boundaries are crossed the various laws come into play, such as Sweden's strict lays about exporting personal information about Swedish citizens.
- Another example is the Canadian law saying that Canadian traffic with source and destination within Canada may not leave the country.

- Two-level routing algorithm needed
  - **Interior gateway protocol used within each network**
  - **Exterior gateway protocol used between networks**
- Thus, one has intranetwork routing and internetwork routing
  • Consider an internet packet starts out from its LAN addressed to the local multiprotocol routine (in the MAC layer header)
    - After the packet has reached the local router, the network layer decides which mutliprotocol router to forward the packet to using its own routing table
    - If that router can be reached using the packet's native network protocol, the packet is directly forward there
    - Otherwise, it is tunnelled there (see previous slide), encapsulated in the protocol required by the intervening network



(a) (b)

Fragmentation

- Each network imposes some maximum size on its packets, due to various causes. Among them: hardware (e.g. the width of a TDM transmission slot), operating system (e.g. buffers of 512 bytes), protocols (e.g. number of bits in the length field), compliance with some standard, desire to reduce error induced retransmissions, desire to prevent packet from occupying the channel too long. Besides starting with packets which are small enough for each intermediate network and gateway, a solution is to allow gateways to break packets up into fragments.

- With **transparent fragmentation**, when an oversized packet arrives at a gateway, it is fragmented and all the packets are send to the same exit gateway of the subnet, where the fragments are recombined. Fragments must be numbered and the last one must be somehow indicated.
- Passage through the small-packet subnet has been make transparent:, subsequent subnets are not aware that this has occurred. ATM networks, for example, have special hardware to provide transparent fragmentation (there called segmentation) of packets into ATM cells and then reassembly of cells into packets.
- With **nontransparent fragmentation**, once a packet is fragmented it is only reassembled when the destination is reached. This requires that every host is able to do reassembly.

The network layer in the Internet

- In the network layer, the Internet can be viewed as a collection of networks or **ASes** (**Autonomous Systems**) that are interconnected.
- There is no real structure, but several major backbones exist. These are constructed from high-bandwidth lines and fast routers.
- The biggest of these back- bones, to which everyone else connects to reach the rest of the Internet, are called **Tier 1 networks**. Attached to the backbones are ISPs (Internet Service Pro- viders) that provide Internet access to homes and businesses, data centers and colocation facilities full of server machines, and regional (mid-level) networks.
- The data centers serve much of the content that is sent over the Internet. Attached to the regional networks are more ISPs, LANs at many universities and companies, and other edge networks.

- The glue that holds Internet together is the network layer protocol **Internet Protocol** (IP). It provides a best-effort way to transport datagrams from source to destination, whether or not these machines are on the same network, or whether or not there are other networks in between them.

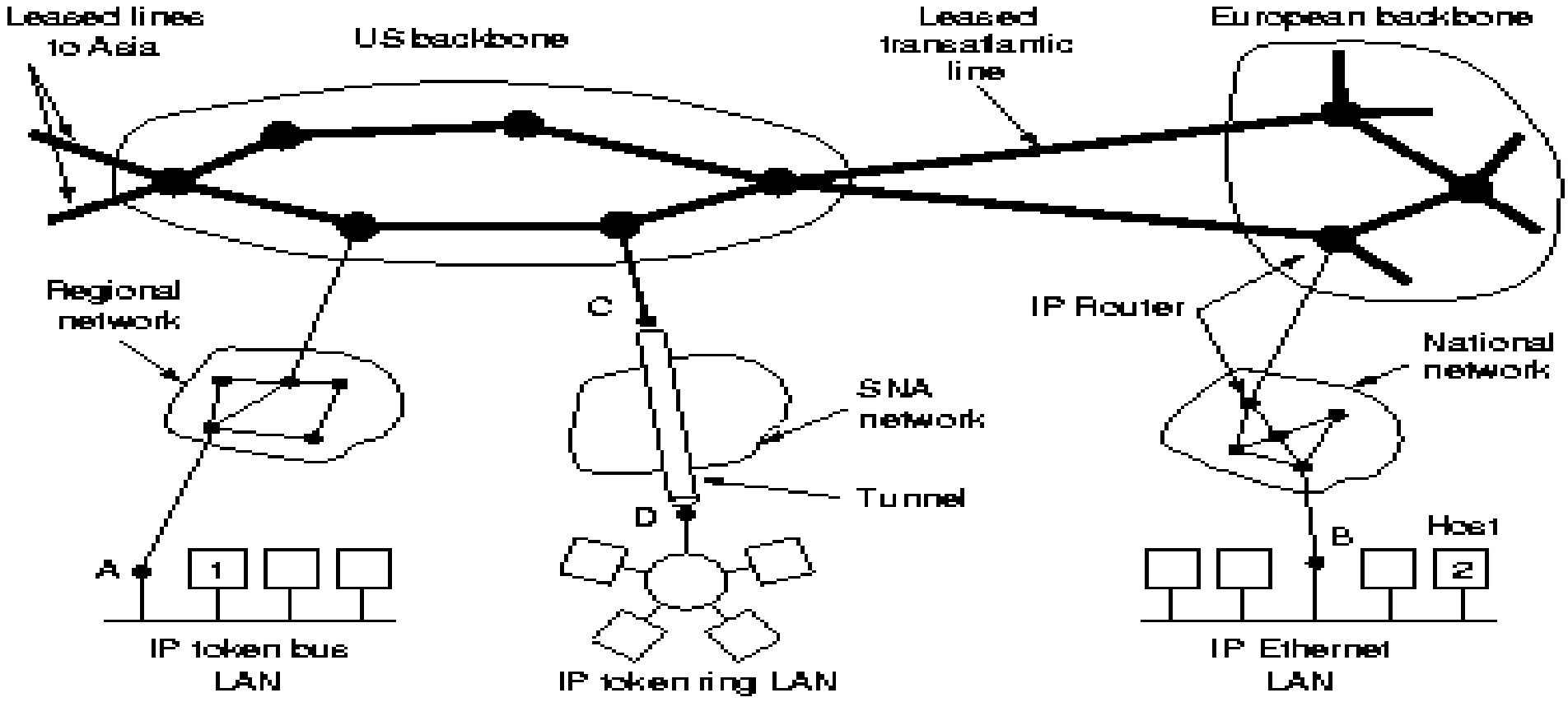


Fig: The Internet is an interconnected collection of many networks.

- In the example of Fig, a packet originating at a host on the home network has to traverse four networks and a large number of IP routers before even getting to the company network on which the destination host is located.
- This is not unusual in practice, and there are many longer paths.
- There is also much redundant connectivity in the Internet, with backbones and ISPs connecting to each other in multiple locations. **This means that there are many possible paths between two hosts. It is the job of the IP routing protocols to decide which paths to use.**

The IP Version 4 Protocol(IPV4)

- An IPv4 datagram consists of a header part and a body or payload part. The header has a 20-byte fixed part and a variable-length optional part.

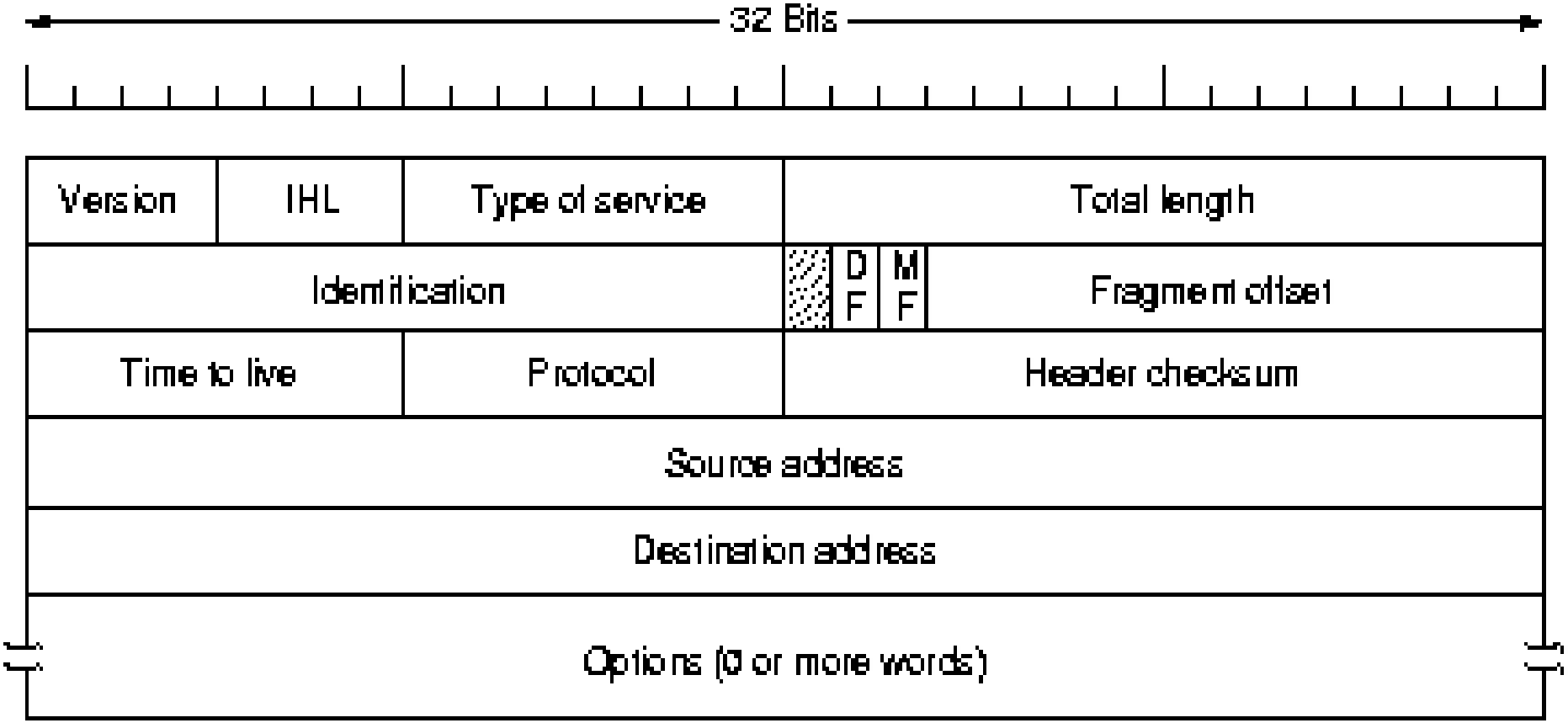


Figure : The IPv4 (Internet Protocol) header.

- **The bits are transmitted from left to right and top to bottom**, with the high-order bit of the ***Version*** **field** going first. (This is a ‘‘big-endian’’ network byte order. On little-endian machines, such as Intel x86 computers, a software conversion. is required on both transmission and reception. The SUN Sparc is big endian, the Pentium is little endian, requiring conversion on both transmission and reception.).This is usually done in hardware.
- **The *Version* field keeps** track of which version of the protocol the datagram belongs to.
- **The IHL field tells** how long the header is, in 32 bit words. The minimum is 5, the maximum is 15, providing for a 40 byte option field.

- The **Type of Service(TOS) field** contains a 3 bit Precedence field, used for the priority from 0 (normal) to 7 (network control packet), and 3 flags Delay, Throughput and Reliability, to specify what is most important for the packet. In practice, current routers mostly ignore the TOS field.
- The **Total Length** includes everything in the datagram, with a maximum length of 65,535 bytes. At present this is tolerable, but with future gigabit networks larger datagrams will be needed.
- The ***Identification*** **field** is needed to allow the destination host to determine which packet a newly arrived fragment belongs to. All the fragments of a packet contain the same *Identification* value.

- Then come two 1-bit fields related to fragmentation. ***DF* stands for Don't Fragment**. It is an order to the routers not to fragment the packet. Originally, it was intended to support hosts incapable of putting the pieces back together again.
- which is the largest packet that can travel along a path without being fragmented. By marking the datagram with the *DF* bit, the sender knows it will either arrive in one piece, or an error message will be returned to the sender.
- ***MF* stands for More Fragments**. All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.

- **The *Fragment offset*** tells where in the current packet this fragment belongs. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit. Since 13 bits are provided, there is a maximum of 8192 fragments per datagram, supporting a maximum packet length up to the limit of the *Total length* field. Working together, the *Identification*, *MF*, and *Fragment offset* fields are used to implement fragmentation.
- The ***TtL (Time to live)* field** is a counter used to limit packet lifetimes. It was originally supposed to count time in seconds, allowing a maximum lifetime of 255 sec. It must be decremented on each hop and is supposed to be decremented multiple times when a packet is queued for a long time in a router.

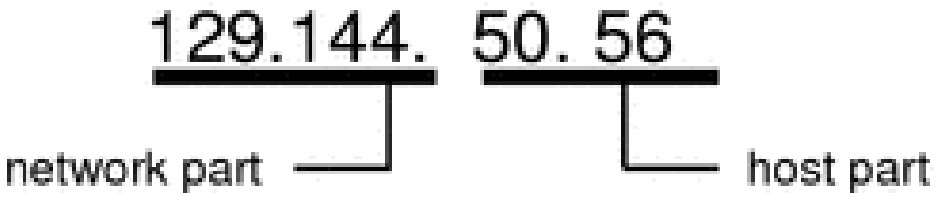
- When the network layer has assembled a complete packet, it needs to know what to do with it. The ***Protocol*** **field tells** it which transport process to give the packet to. TCP is one possibility, but so are UDP and some others. The numbering of protocols is global across the entire Internet.
- The **Header checksum** verifies the header only, useful for detecting errors by bad memory bytes or corrupted software inside a router.
- **The *Source address* and *Destination address*** indicate the IP address of the source and destination network interfaces.

- The ***Options* field** was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design, The options are of variable length. Each begins with a 1-byte code identifying the option. Some options are followed by a 1-byte option length field, and then one or more data bytes.

- Each network that runs TCP/IP must have a unique network number. Every machine on the network must have a unique IP address. You must understand how IP addresses are constructed before you register your network and obtain its network number
- The IPv4 address is a 32-bit number that uniquely identifies a network interface on a machine. An IPv4 address is typically written in decimal digits, formatted as four 8-bit fields that are separated by periods. Each 8-bit field represents a byte of the IPv4 address. This form of representing the bytes of an IPv4 address is often referred to as the **dotted-decimal format**.

- The bytes of the IPv4 address are further classified into two parts: the network part and the host part. The following figure shows the component parts of a typical IPv4 address, 129.144.50.56.

Parts of an IPv4 Address



Network Part

- The network part specifies the unique number that is assigned to your network. The network part also identifies the class of network that is assigned. In Figure, the network part occupies two bytes of the IPv4 address.

Host Part

- This is the part of the IPv4 address that you assign to each host. The host part uniquely identifies this machine on your network.
- Note that for each host on your network, the network part of the address is the same, but the host part must be different.

IP addresses

- A defining feature of IPv4 is its 32-bit addresses. Every host and router on the Internet has an IP address that can be used in the *Source address* and *Destination address* fields of IP packets.
- IP addresses are hierarchical, unlike Ethernet addresses. Each 32-bit address is comprised of a variable-length network portion in the top bits and a host portion in the bottom bits. The network portion has the same value for all hosts on a single network, such as an Ethernet LAN. This means that a network corresponds to a contiguous block of IP address space. This block is called a **prefix**.

- IP addresses are written in **dotted decimal notation**. In this format, each of the 4 bytes is written in decimal, from 0 to 255. For example, the 32-bit hexadecimal address 80D00297 is written as 128.208.2.151.
- The IP address 0.0.0.0 is used by hosts when they are being booted but is not used afterwards.

- IP addresses with 0 as network number refer to the current network. The address 255.255.255.255 allows broadcasting on the local network, typically a LAN. The addresses with a proper network number and all 1's in the host field, allow a broadcast on the distant network.
- All addresses 127.x.y.z are for loopback testing, packets sent to that address are not put on the wire, but just put in the input queue of the same machine. This allows for debugging network software.

Internet control protocols

- In addition to IP, which is used for data transfer, the Internet has several control protocols.

Internet Control Message Protocol (ICMP)

- When something unexpected occurs in a router or host, this event is reported by ICMP. It is also used by routers to test the internet or to obtain information to be use in routing decisions. The most important message types are given in this table. Each ICMP message is encapsulated in an IP packet.

- Routers and other network devices monitor the operation of the network. When an error occurs, these devices send a message using ICMP. Messages that can be sent include "destination unreachable", "time exceeded", and "echo requests".
  - ICMP is a network layer protocol.
  - ICMP messages are not passed directly to the data link layer. The message is first encapsulated inside the IP datagram before going to the lower layer.

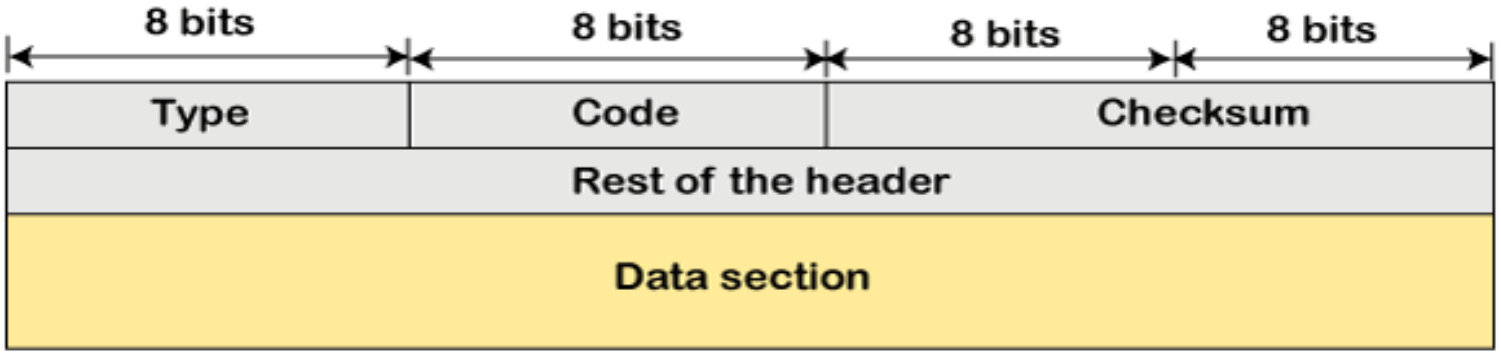
- **The ICMP messages are usually divided into two categories:**

ICMP messages

Category	Type	Message
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

- **Information Messages** − In this message, the sender sends a query to the host or router and expects an answer. For example, A host wants to know if a router is alive or not.
- **Error-reporting message** − This message report problems that a router or a host (destination) may encounter when it processes an IP packet.
- **Query Message** − It helps a router or a network manager to get specific information from a router or another host.

- **The ICMP message contains the following fields:**



Type: It is an 8-bit field. It defines the ICMP message type. The values range from 0 to 127 are defined for ICMPv6, and the values from 128 to 255 are the informational messages.
Code: It is an 8-bit field that defines the subtype of the ICMP message
Checksum: It is a 16-bit field to detect whether the error exists in the message or not.

IPV6 internet protocol

- The **main difference** between IPv4 and IPv6 Header is that **the source and destination addresses of IPv4 header are 32 bit long while the source and destination addresses of IPv6 header are 128 bits long.**

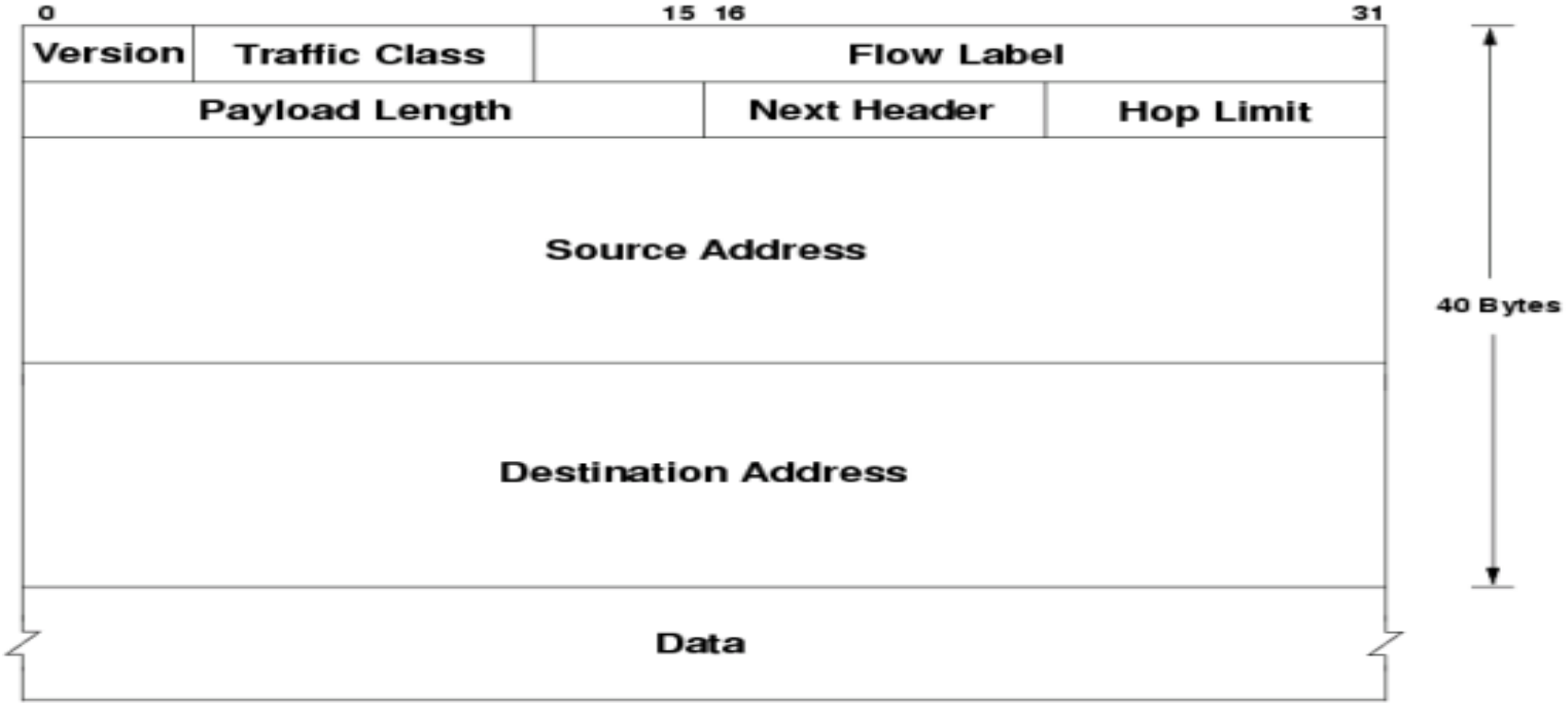


Fig: The IPv6 fixed header (required).

- **Version** – It is a 4-bit field that describes the IP type that is being used.
- **Traffic class** (*Differentiated services):* it is an 8-bit field that describes the packet's class or priority. Moreover, it is similar to the IPv4 ToS field.
- **Flow label** – It is 20-bit long. Moreover, it indicates the position of the packet in a set of packets and helps to prioritize the packets, especially when transmitting voice.
- **Payload length** – It is 16-bit long and displays the length of IPv6 payload with the extension headers and upper layer protocol data.

- **Next Header** – It is an 8-bit field that shows the type of the first extension or the protocol in the upper layer.
- **Hop Limit** – It is 8-bit long. It indicates the maximum number of routers the packet is capable of passing. This is similar to TTL field in IPv4 header.
- **Source address** – It is 128 bits long. It is the address of the device that sends the packet.
- **Destination address** – It is also 128 bits long. It is the address of the device that receives the packet.
- **Data** – It represents the real data that should be transmitted.

- A new notation has been devised for writing 16-byte addresses. They are written as eight groups of four hexadecimal digits with colons between the groups, like this:

  8000:0000:0000:0000:0123:4567:89AB:CDEF

- Since many addresses will have many zeros inside them, three optimizations have been authorized. First, leading zeros within a group can be omitted, so 0123 can be written as 123. Second, one or more groups of 16 zero bits can be replaced by a pair of colons. Thus, the above address now becomes

  8000::123:4567:89AB:CDEF

- Finally, IPv4 addresses can be written as a pair of colons and an old dotted decimal number, for example:

  ::192.31.20.46

Features of IPV6

There are various features of IPV6, which are as follows−

- **Larger address space:** An IPV6 address is 128 bits long. It is compared with the 32-bit address of IPV4. It will allow for unique IP-addresses up to 3.4×10^{38} whereas IPV4 allows up to 4.3×10^{8} unique address.
- **Better Header format:** New header form has been designed to reduce overhead. It is done by moving both non-essential fields and optional fields to extension field header that are placed after the IPV6 header.
- **More Functionality:** It is designed with more options like priority of packet for control of congestion, Authentication etc.
- **Allowance for Extension:** It is designed to allow the extension of the protocol if required by new technologies.
- **Support of resource allocation:** In IPV6, the type of service fields has been removed, but a new mechanism has been added to support traffic control or flow labels like real-time audio and video.